



香港科技大學
THE HONG KONG UNIVERSITY OF SCIENCE AND TECHNOLOGY

ELEC 2100 Signals and Systems

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Outline Chapter 4 Frequency domain analysis of LTIS



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AND TECHNOLOGY

1 Overview

2 Frequency Response of System

3 Amplitude Modulation

4 Sampling Theorem and Aliasing

5 Fourier Transform of a Sequence (DTFT)

6 Discrete Time Fourier Series

7 DTFT and DFT

8 Fast Fourier Transform (Reference)

1. Overview of Frequency Domain Analysis OF LTIS

This chapter preliminarily studies several aspects of **applying Fourier transform method to communication systems**

Fourier Transform Property	Application
Time-domain Convolution Theorem	Filtering , System Frequency Response Characteristics
Frequency Shift Property	Modulation , Frequency Division Multiplexing
Frequency-Domain Convolution Theorem	Sampling , Time Division Multiplexing

1. Overview of DTFT

From **Continuous-Time** Fourier Transform (CTFT) to **Discrete-Time** Fourier Transform (DTFT)

- Discrete Time Fourier Series (DTFS)
- Properties of Discrete Time Fourier Series
- Discrete Time Fourier Transform (DTFT)
- Properties of Discrete Time Fourier Transform
- Fast Fourier Transformer (FFT)

2. Frequency Response of System

Main Content

Fourier Transform Analysis Method:

- System Function in the Form of Fourier Transform
- Ideal Low Pass Filter

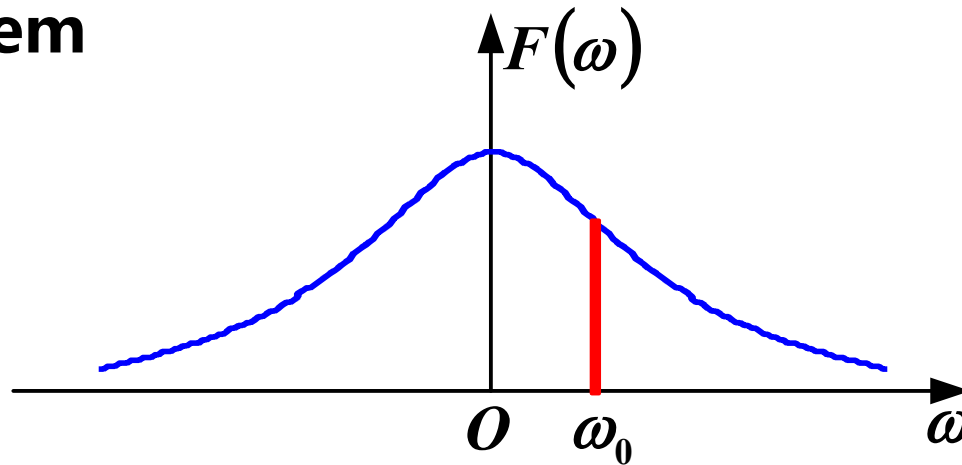
2. Frequency Response of System

1. System Function in the Form of Fourier Transform

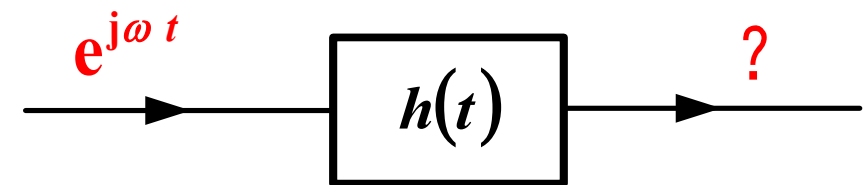
Decomposition: A signal can be decomposed into fundamental signals $e^{j\omega t}$

Response Calculation: Determine the response of the fundamental signal $e^{j\omega t}$ passing through the system

Superposition: The response of the signal passing through the system

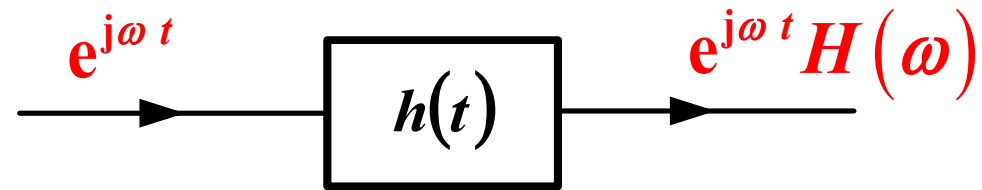


$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{j\omega t} d\omega$$



2. Frequency Response of System

(1) Response of the fundamental signal $e^{j\omega t}$ passing through the system



System Response

$$\begin{aligned} r_1(t) &= e^{j\omega t} * h(t) = \int_{-\infty}^{\infty} h(\tau) e^{j\omega(t-\tau)} d\tau \\ &= e^{j\omega t} \int_{-\infty}^{\infty} h(\tau) e^{-j\omega\tau} d\tau \\ &= e^{j\omega t} H(\omega) \end{aligned}$$

Where

$$H(\omega) = \mathcal{F}[h(t)] = \int_{-\infty}^{\infty} h(\tau) e^{-j\omega\tau} d\tau$$

2. Frequency Response of System

(2) Signals can be expressed as a sum of fundamental signals

Periodic Signals

$$e(t) = \sum_{k=-\infty}^{\infty} E_k e^{jk\omega_1 t}$$

Fundamental Signal $e^{jk\omega_1 t}$

Aperiodic Signals

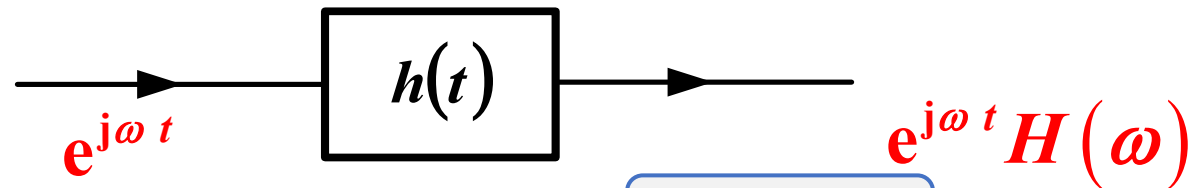
$$e(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} E(\omega) e^{j\omega t} d\omega$$

Fundamental Signal $e^{j\omega t}$

2. Frequency Response of System

(3) Response of an arbitrary signal through a system

Method: Decompose the signal, determine individual responses, then superimpose



Component Response:

$$\frac{1}{2\pi} E(\omega) d\omega e^{j\omega t}$$

$$\frac{1}{2\pi} E(\omega) d\omega e^{j\omega t} H(\omega)$$

Homogeneity

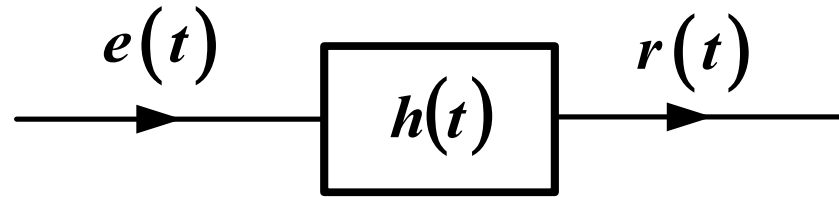
$$e(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} E(\omega) e^{j\omega t} d\omega \quad r(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} E(\omega) e^{j\omega t} H(\omega) d\omega$$

Additivity

$$R(\omega) = E(\omega) \cdot H(\omega)$$

2. Frequency Response of System

(3) Response of an arbitrary signal through a system



System Response

$$r(t) = e(t) * h(t)$$

Let

$$\mathcal{F}[e(t)] = E(\omega), \mathcal{F}[h(t)] = H(\omega), \mathcal{F}[r(t)] = R(\omega)$$

Using the time-domain convolution theorem, we obtain

$$R(\omega) = E(\omega) \cdot H(\omega)$$

2. Frequency Response of System

(4) System function in the Fourier transform domain

$$H(\omega) = \frac{R(\omega)}{E(\omega)} \quad \text{or } H(j\omega)$$

$H(\omega)$ is referred to as the **system function** in the Fourier transform domain

$H(\omega) \sim \omega$ is also termed the **frequency response characteristic** of the system

$$H(\omega) = |H(\omega)| e^{j\varphi(\omega)}$$

$|H(\omega)| \sim \omega$ **Magnitude-Frequency Characteristic**

$\varphi_h(\omega) \sim \omega$ **Phase-Frequency Characteristic**

2. Frequency Response of System

(5) Physical interpretations of the system function

	$R(\omega) = E(\omega) \cdot H(\omega)$	
Excitation	$E(\omega) = E(\omega) \cdot e^{j\varphi_e(\omega)}$	frequency-dependent weighting on the signal's spectral components
System Function	$H(\omega) = H(\omega) \cdot e^{j\varphi_h(\omega)}$	
Response	$R(\omega) = R(\omega) \cdot e^{j\varphi_r(\omega)}$	

Through complex function operations, we obtain

$$|R(\omega)| = |E(\omega)| \cdot |H(\omega)|$$

The amplitude $E(\omega)$
is scaled by $|H(\omega)|$

$$\varphi_r(\omega) = \varphi_e(\omega) + \varphi_h(\omega)$$

The phase of $E(\omega)$
is shifted by $\varphi_h(\omega)$

2. Frequency Response of System

(6) Methods for determining the system function

- **From Differential Equations**
- **From Circuit Analysis**
- **From Impulse Response**

2. Frequency Response of System

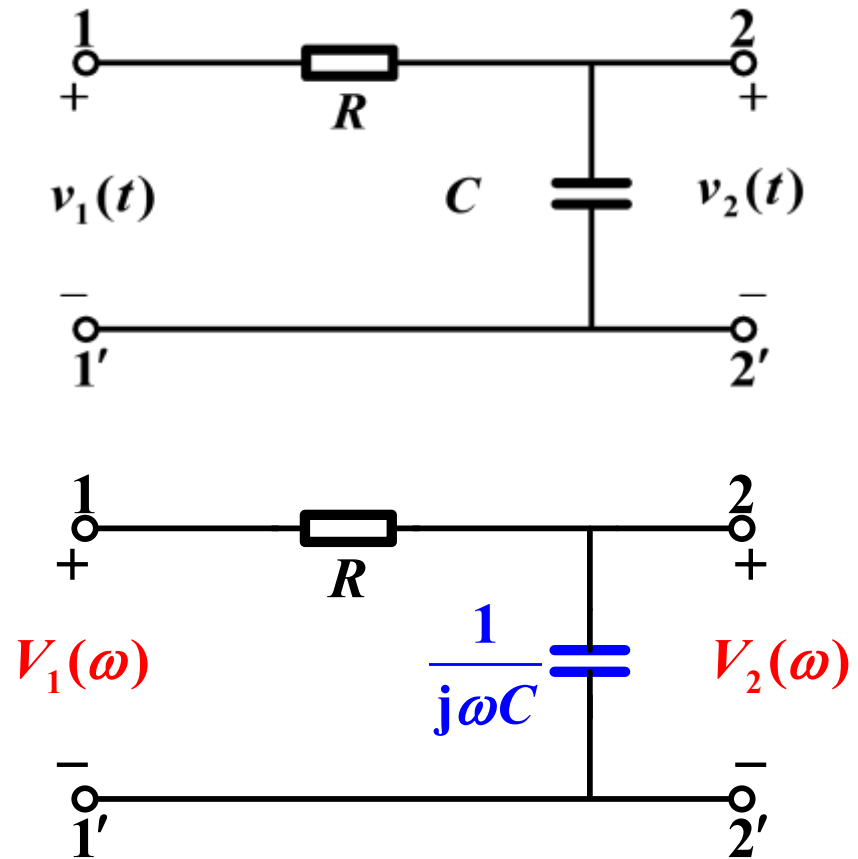
Example 1

Determine the system function $H(\omega)$ for the given RC circuit

Represent the circuit in the frequency domain and derive the system function

$$H(\omega) = \frac{V_2(\omega)}{V_1(\omega)} = \frac{\frac{1}{j\omega C}}{R + \frac{1}{j\omega C}}$$

Let $\alpha = \frac{1}{RC}$, $H(\omega) = \frac{\alpha}{\alpha + j\omega}$



2. Frequency Response of System

Example 1

Frequency Response Characteristics of the System

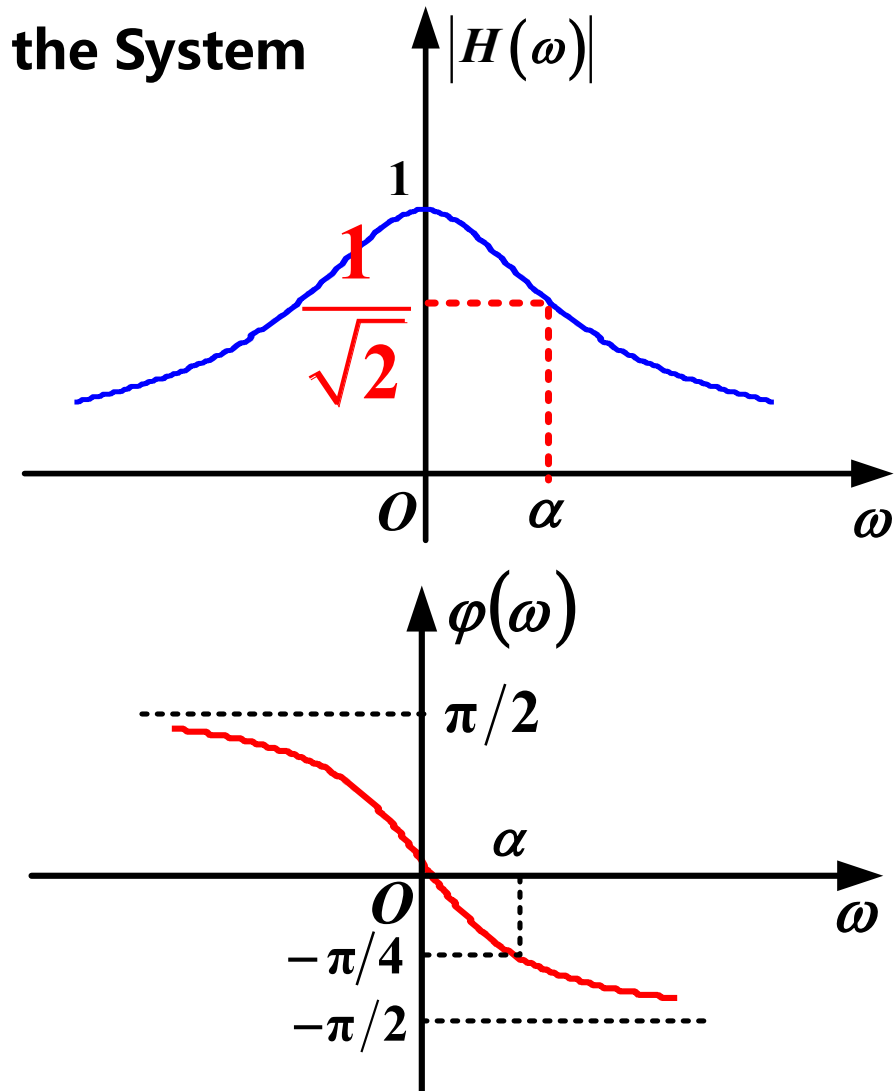
$$H(\omega) = \frac{\alpha}{\alpha + \mathbf{j}\omega} \quad \left(\alpha = \frac{1}{RC} \right)$$

$$|H(\omega)| = \frac{\alpha}{\sqrt{\alpha^2 + \omega^2}}$$

$$\varphi(\omega) = -\arctan \frac{\omega}{\alpha}$$

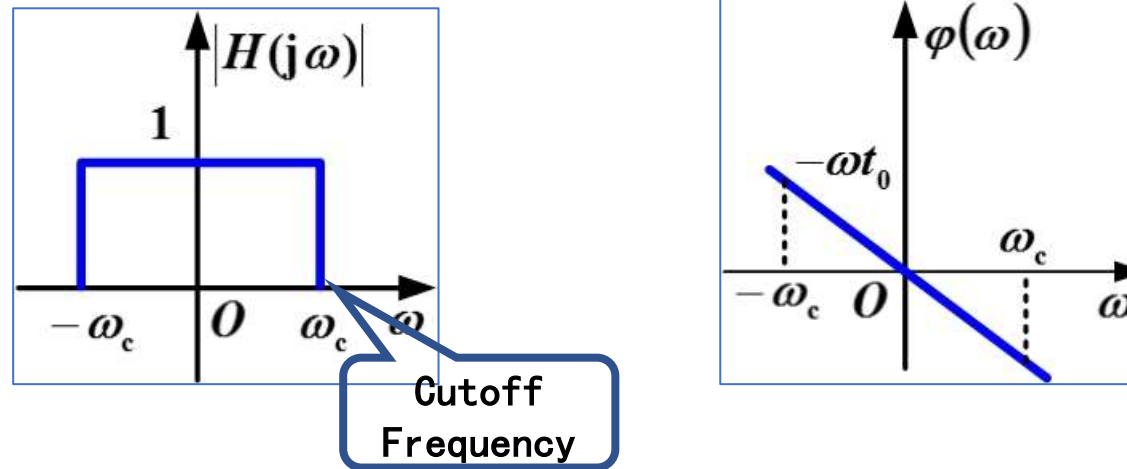
Low-Pass Filter

The **cutoff frequency** is determined by α
 α corresponds to the **3dB** bandwidth
(frequency at which power is halved)



2. Frequency Response of System

2. Ideal Low Pass Filter—Frequency Domain Characteristics



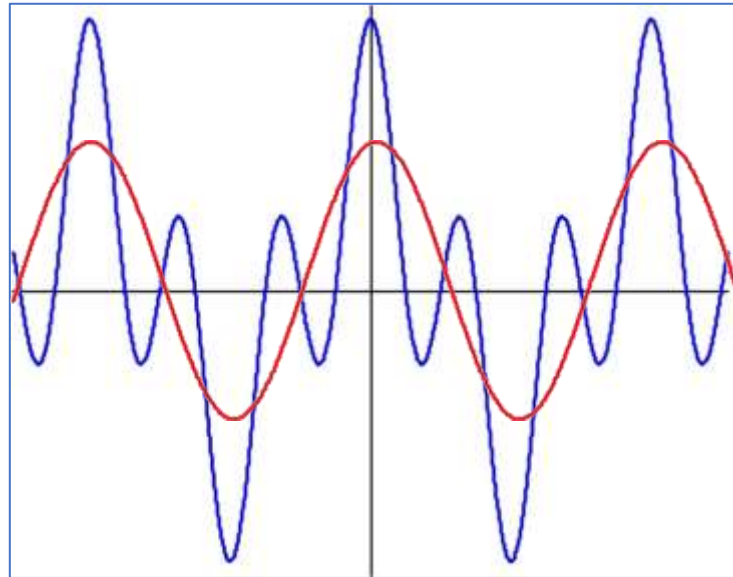
$$H(j\omega) = \begin{cases} 1 \cdot e^{-j\omega t_0} & |\omega| < \omega_c \\ 0 & |\omega| > \omega_c \end{cases} \Rightarrow \begin{cases} |H(j\omega)| = \begin{cases} 1 & |\omega| < \omega_c \\ 0 & |\omega| > \omega_c \end{cases} \\ \phi(\omega) = -\omega t_0 \end{cases}$$

2. Frequency Response of System

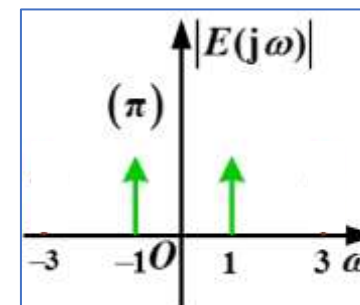
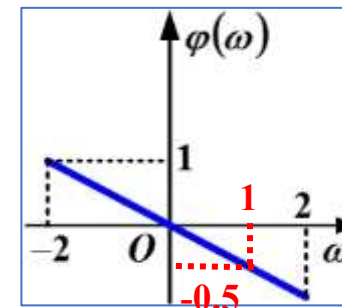
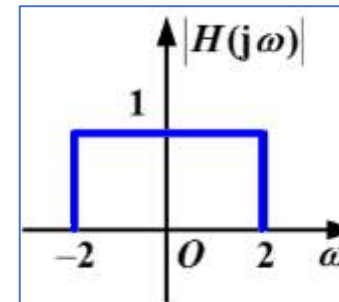
Example 2

An input signal $e(t) = \cos t + \cos 3t$ passes through an ideal low-pass filter $H(j\omega)$

Determine the output signal $r(t)$



$$r(t) = \cos\left(t - \frac{1}{2}\right)$$



2. Frequency Response of System

Summary

$$H(\omega) = \frac{R(\omega)}{E(\omega)}$$

- $H(\omega)$ **System Function in the Form of Fourier Transform**

$$|R(\omega)| = |E(\omega)| \cdot |H(\omega)|$$

The amplitude $E(\omega)$
is scaled by $|H(\omega)|$

$$\varphi_r(\omega) = \varphi_e(\omega) + \varphi_h(\omega)$$

The phase of $E(\omega)$
is shifted by $\varphi_h(\omega)$

- **Ideal Low Pass Filter**

Main Content

- **Modulation and Demodulation**
- **Introduction to Modulation Methods**
- **Amplitude Modulation**
- **Demodulation**
- **Frequency Division Multiplexing**
- **Time Division Multiplexing**

1. Modulation and Demodulation

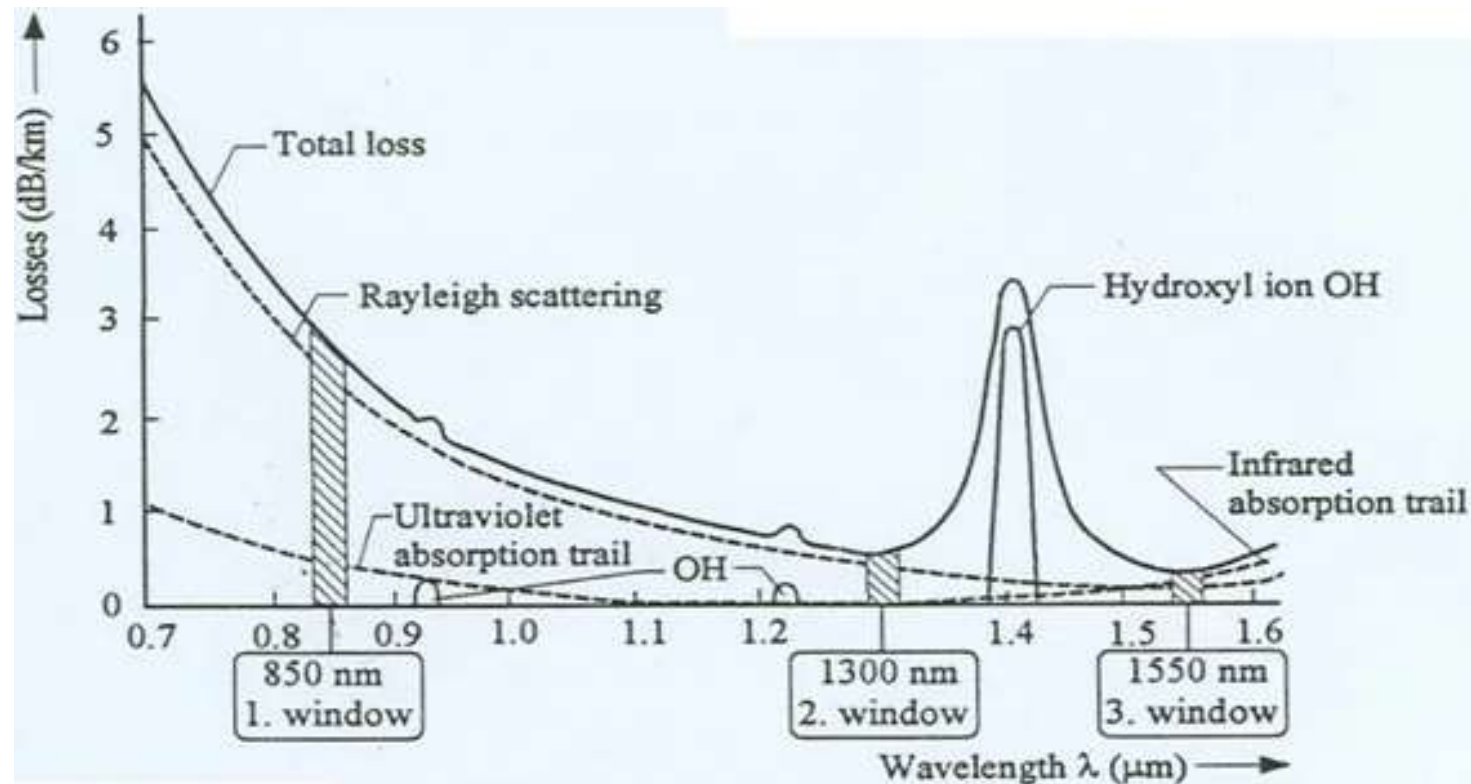
In a communication system, signals are transmitted from the transmitter to the receiver. To achieve signal transmission, **modulation and demodulation** are often required

- **Modulation:** The process of embedding an information-carrying signal into another signal
- **Demodulation:** The process of extracting the information-carrying signal from the modulated signal

3. Amplitude Modulation

(1) Why is Modulation Necessary?

To adapt the signal for transmission over a suitable channel



Fiber loss and working window

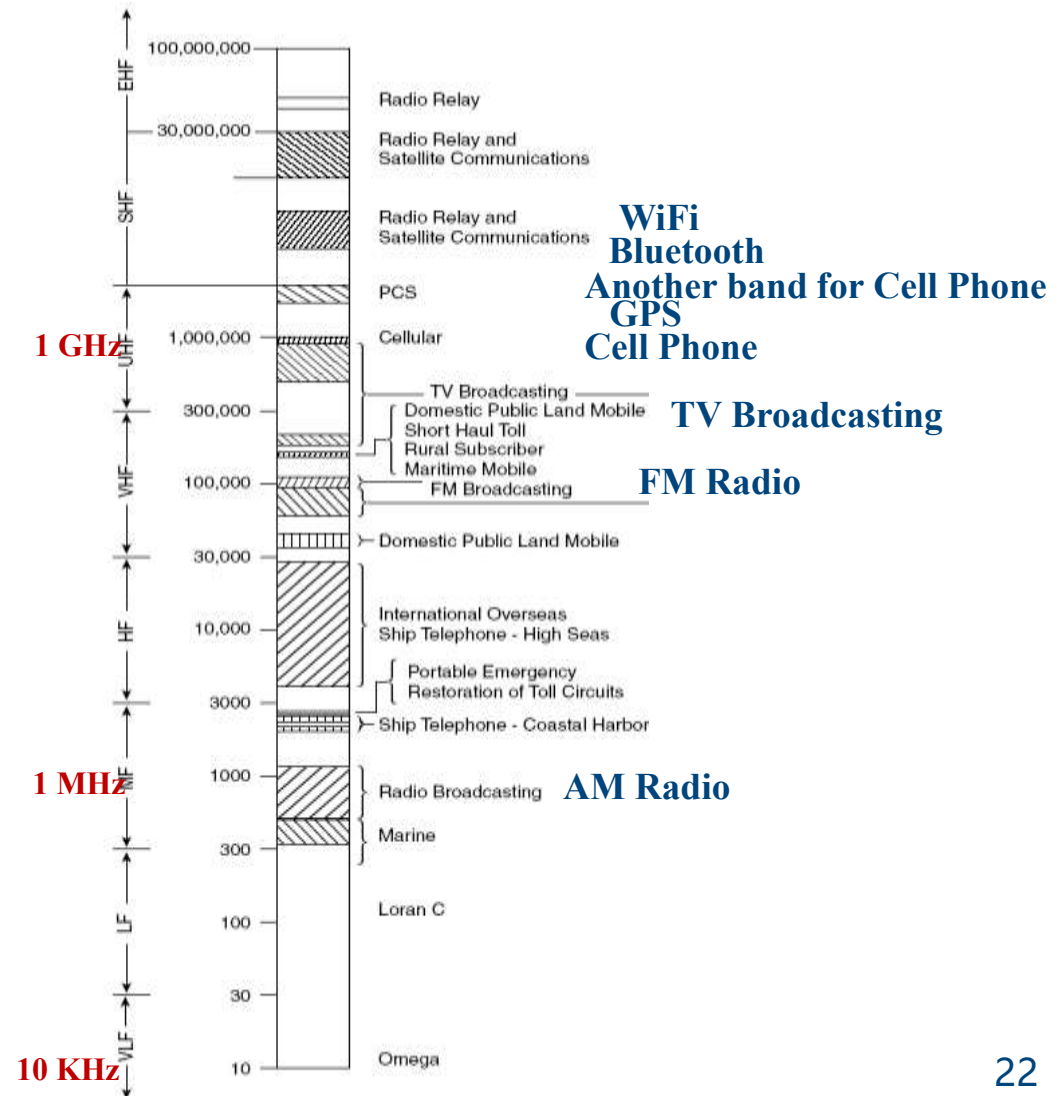
3. Amplitude Modulation

(1) Why is Modulation Necessary?

For about 120 years, human have used radio waves in the range of 10 KHz to 100 GHz for many communications applications other than AM and FM radio broadcasting: marine and aviation communications, terrestrial TV, Satellite TV, walkie talkie, cordless phone, car key, garage door opener, cell phone (2G, 3G, 4G, 5G), WiFi, Bluetooth, RFID, GPS..... etc.

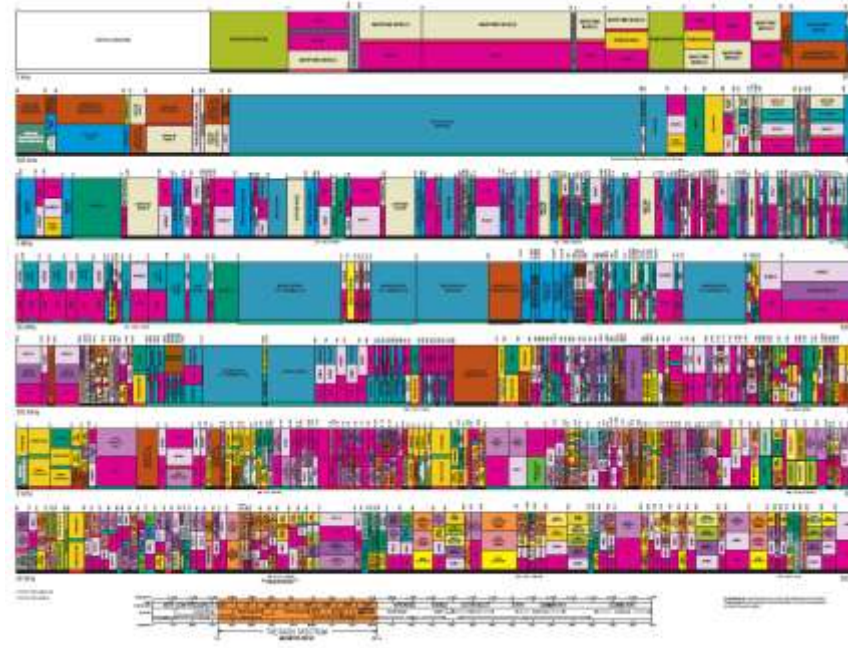
Radio spectrum is a valuable resource with usage governed by governments and international treaties. To the right is a high-level view of how the radio spectrum is allocated to different applications in the US.

Often one has to pay large sum of \$\$ for the use of specific spectrums – spectrum auction.



3. Amplitude Modulation

(1) Why is Modulation Necessary?

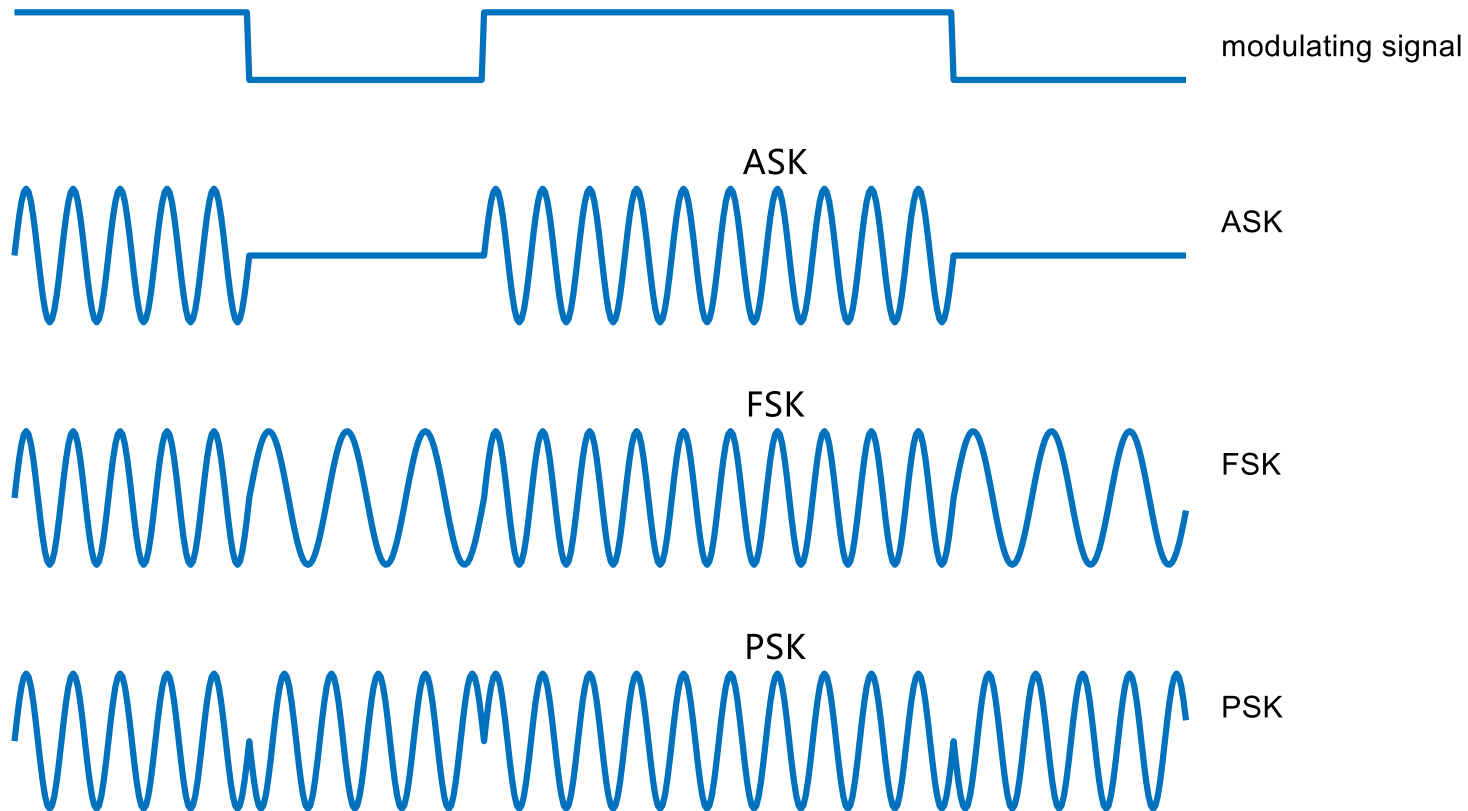


- The detailed frequency allocation table for different applications is extremely complex. It also varies from country to country
- Further, each application may have several to hundreds of channels (e.g., different TV/radio channels) sharing the allocated spectrum
- **Today, every bit of the usable radio spectrum is used!** Because spectrum is so valuable, different applications and different channels are often closely packed with little room between them. (*Hence we need good LPF or BPF to separate different channels.*)

3. Amplitude Modulation

2. Introduction to Modulation Methods

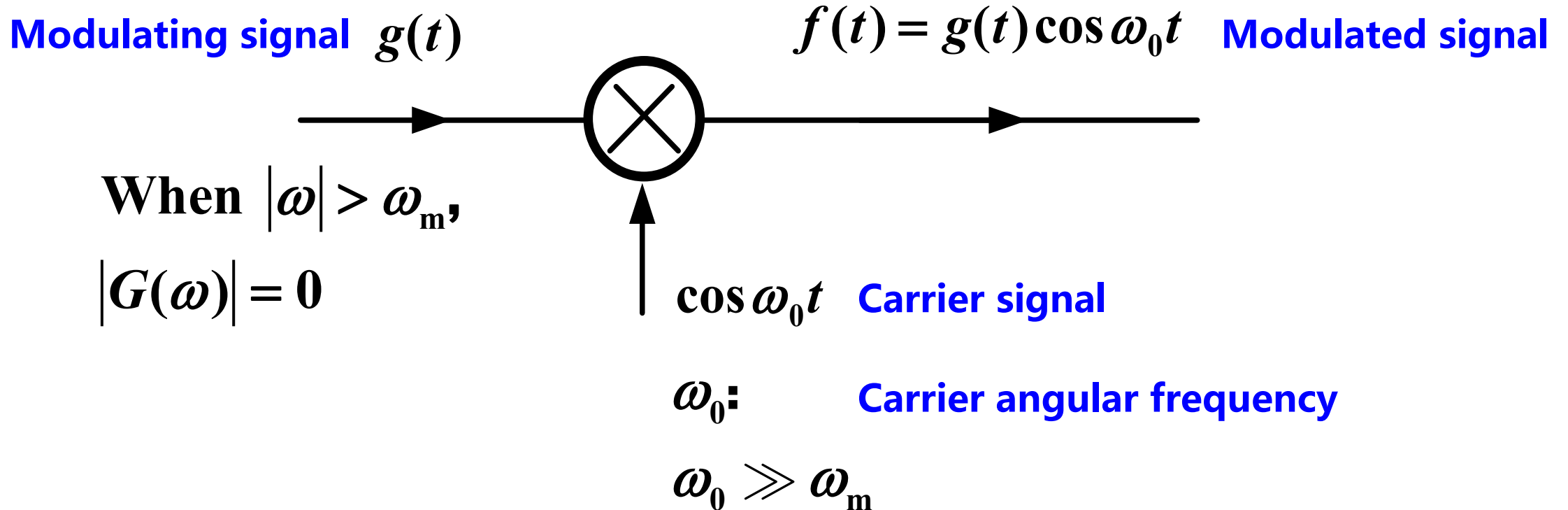
$$f(t) = A \sin(2\pi f t + \theta)$$



3. Amplitude Modulation

3. Amplitude Modulation (Suppressed-carrier amplitude modulation, AM-SC)

(1) System Block Diagram



(2) Spectral Structure

$$f(t) = g(t) \cdot \cos(\omega_0 t) \quad \underline{\text{Euler's formula}} \quad \frac{1}{2} g(t) \left[e^{j\omega_0 t} + e^{-j\omega_0 t} \right]$$

Modulation
theorem

$$F(\omega) = \frac{1}{2} \left[G(\omega - \omega_0) + G(\omega + \omega_0) \right]$$

$$f(t) = g(t) \cos(\omega_0 t)$$

$$\xleftarrow{\text{Convolution theorem}} \frac{1}{2\pi} G(\omega) * \left[\pi \delta(\omega - \omega_0) + \pi \delta(\omega + \omega_0) \right]$$

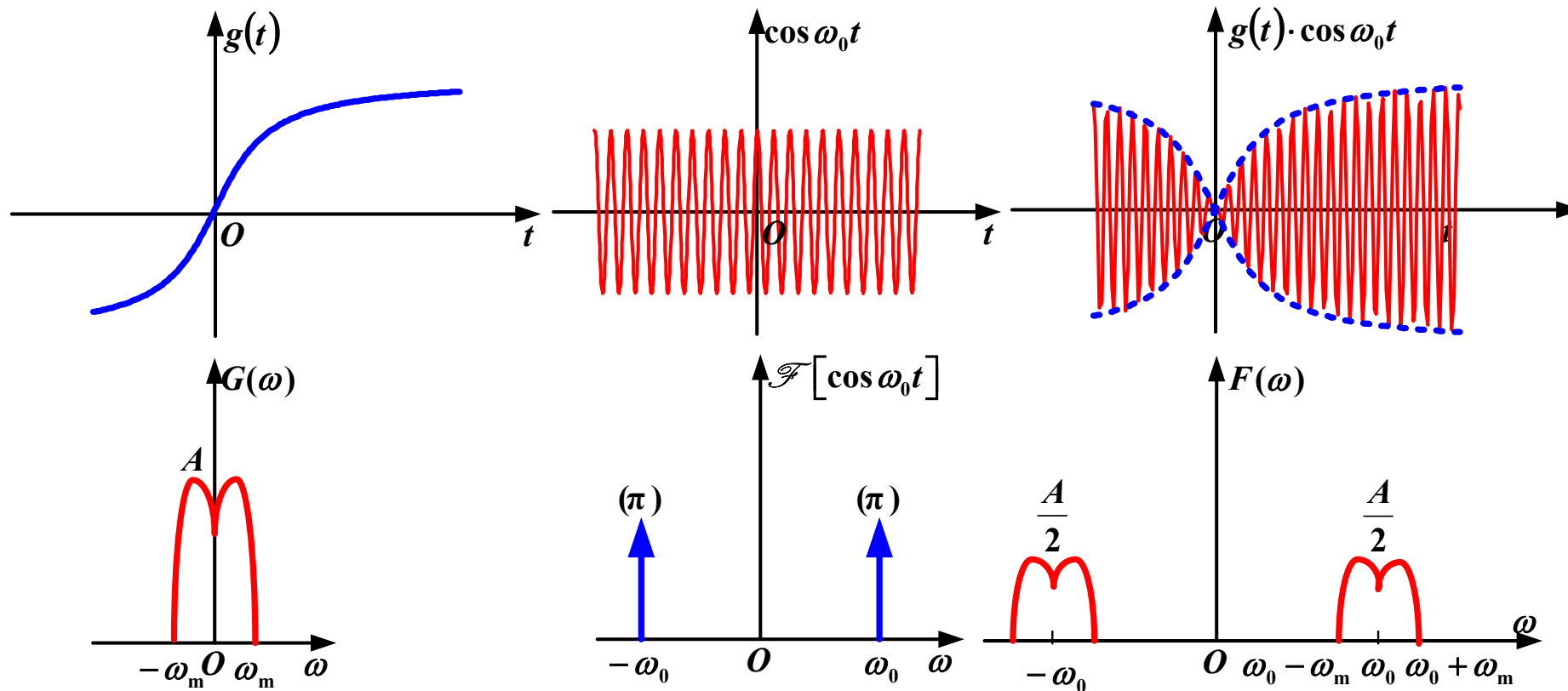
Frequency Shift

Convolution

3. Amplitude Modulation

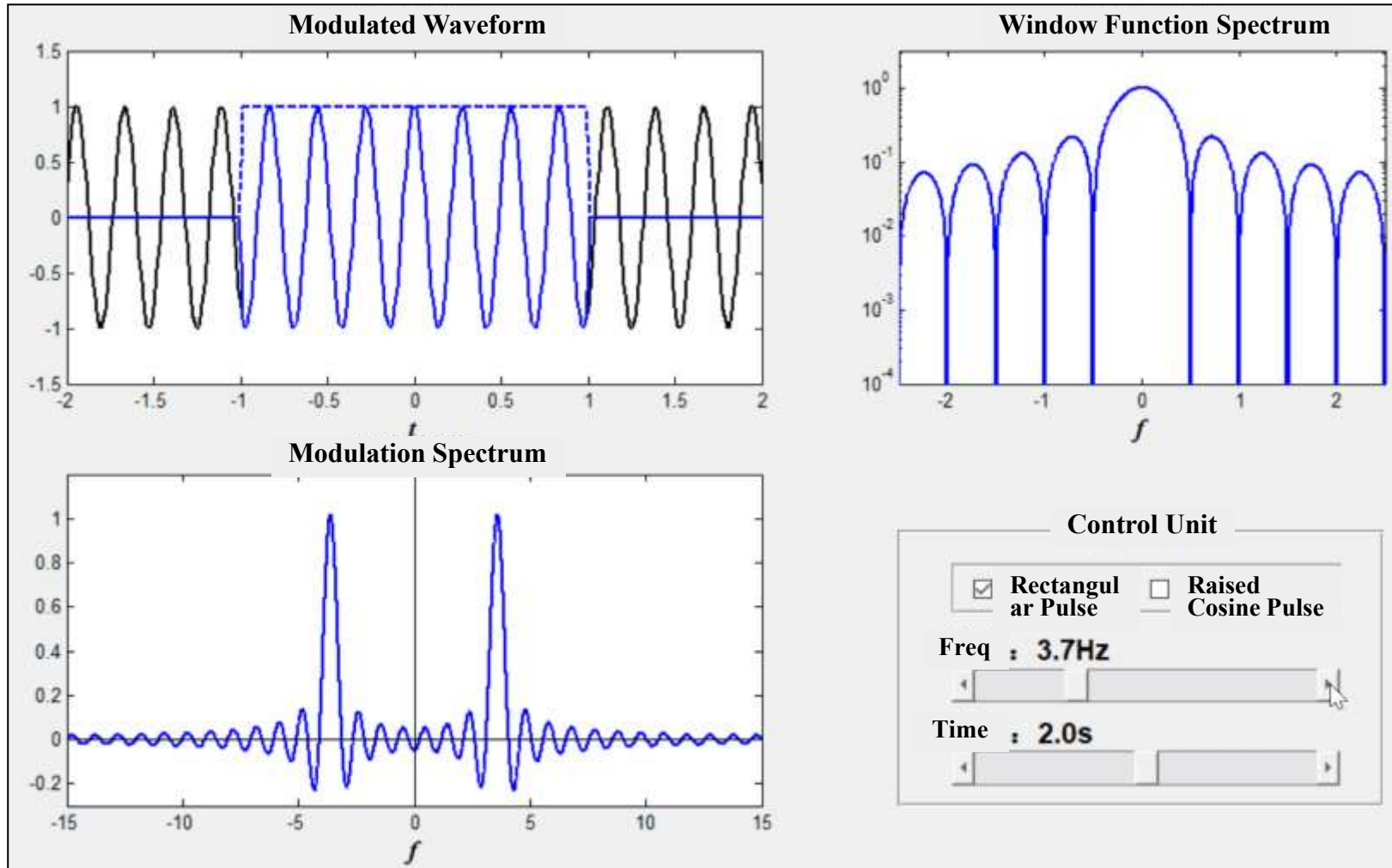
(2) Spectral Structure

$$f(t) = g(t) \cdot \cos(\omega_0 t) \quad F(\omega) = \frac{1}{2} [G(\omega - \omega_0) + G(\omega + \omega_0)]$$



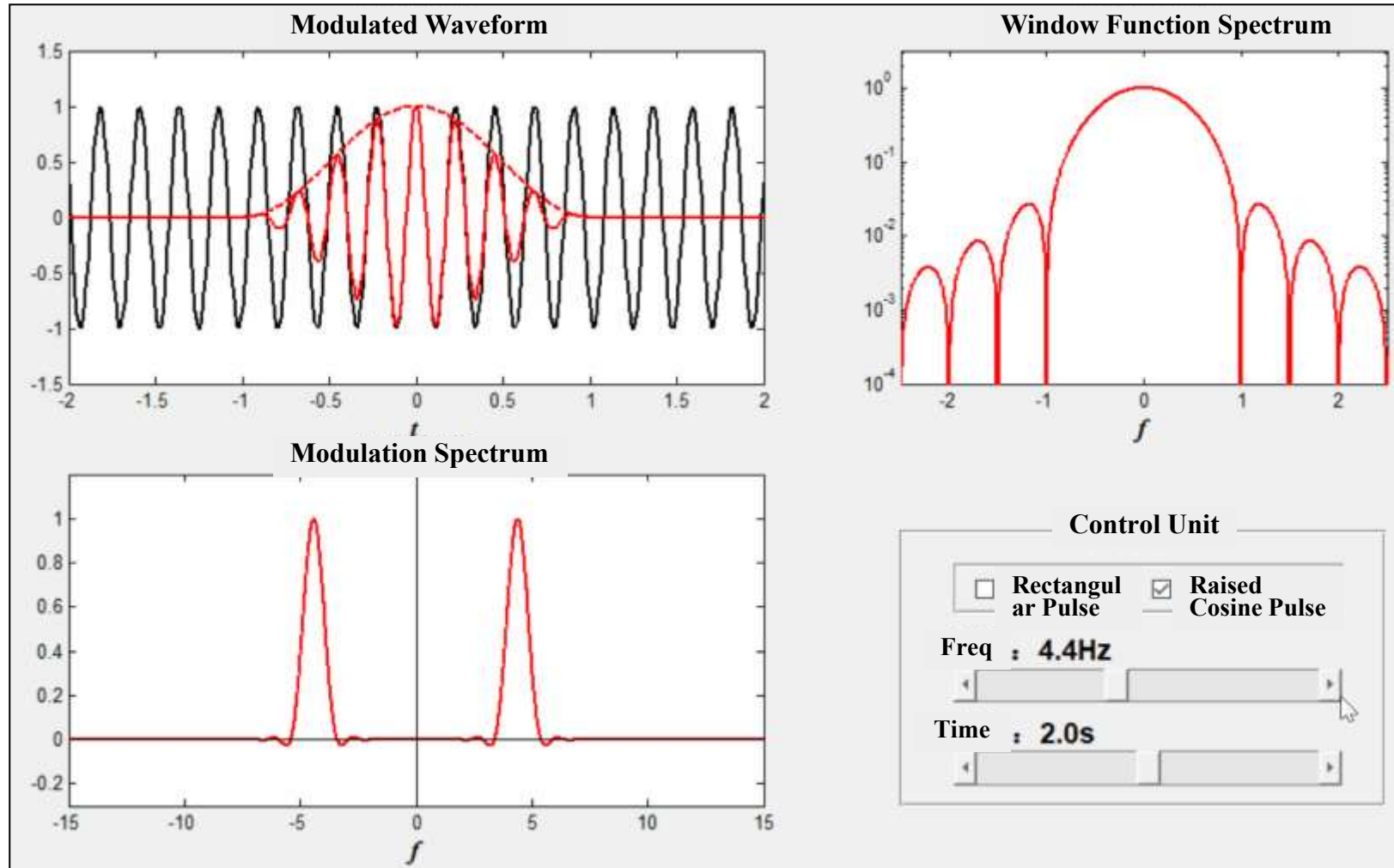
3. Amplitude Modulation

(2) Spectral Structure



3. Amplitude Modulation

(2) Spectral Structure

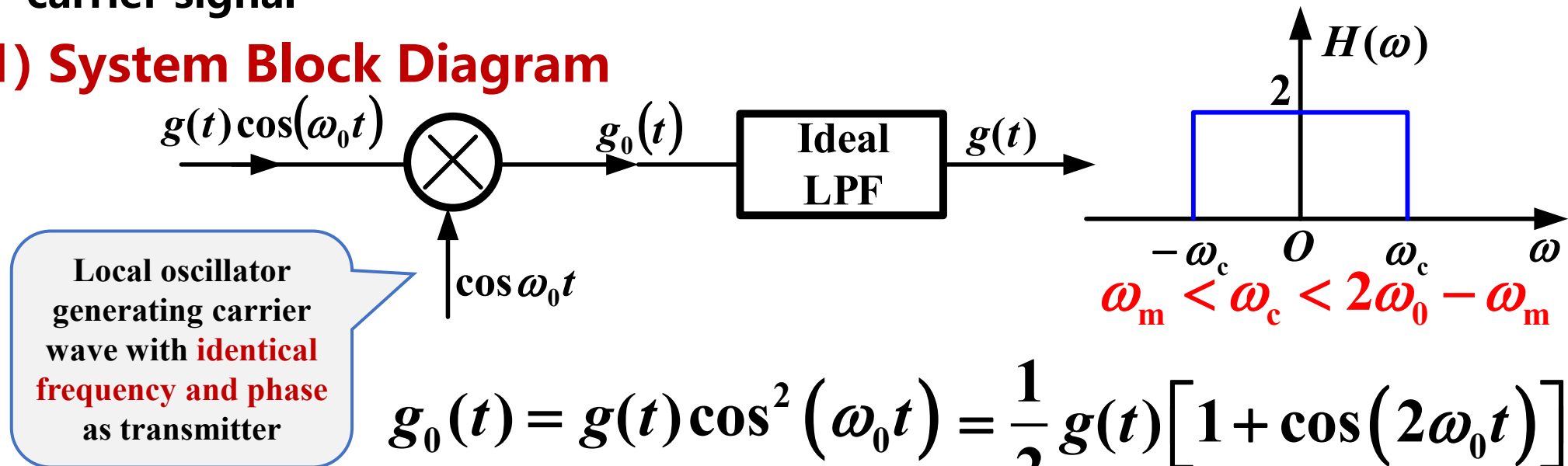


3. Amplitude Modulation

4. Demodulation (Coherent Detection/Synchronous Detection)

The process of recovering the original baseband signal from the modulated carrier signal

(1) System Block Diagram

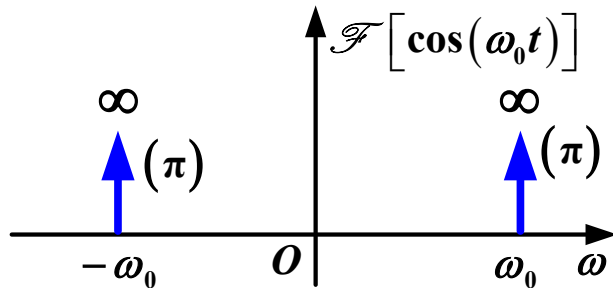
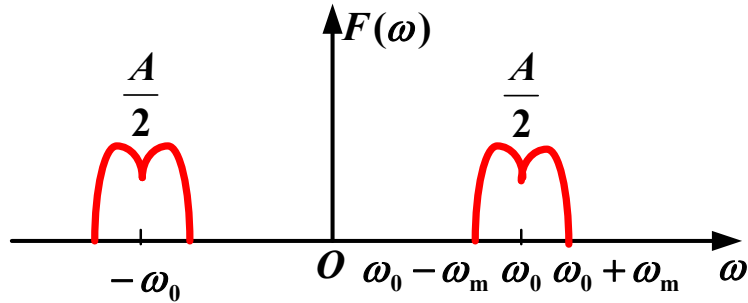


$$g_0(t) = g(t) \cos^2(\omega_0 t) = \frac{1}{2} g(t) [1 + \cos(2\omega_0 t)]$$

$$G_0(\omega) = \frac{1}{2} G(\omega) + \frac{1}{4} G(\omega - 2\omega_0) + \frac{1}{4} G(\omega + 2\omega_0)$$

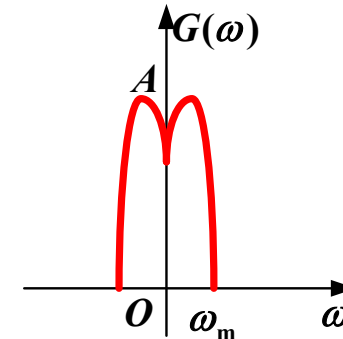
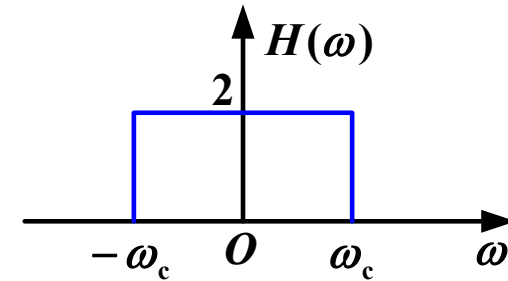
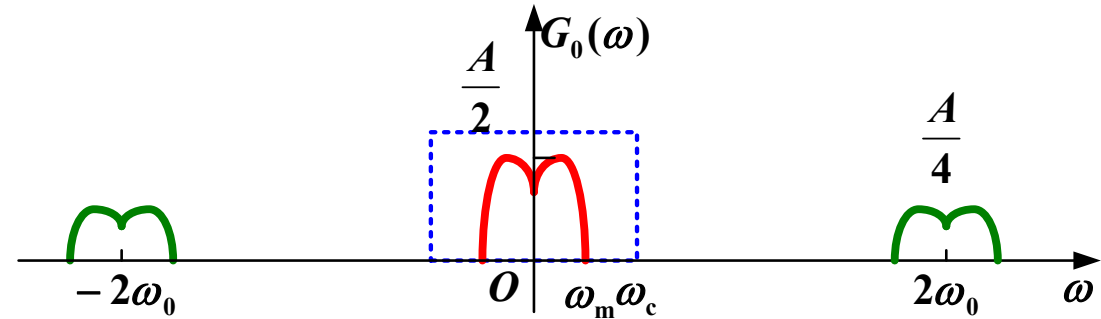
3. Amplitude Modulation

(2) Spectral Structure



$$G_0(\omega)H(\omega) = G(\omega)$$

$$\omega_m < \omega_c < 2\omega_0 - \omega_m$$



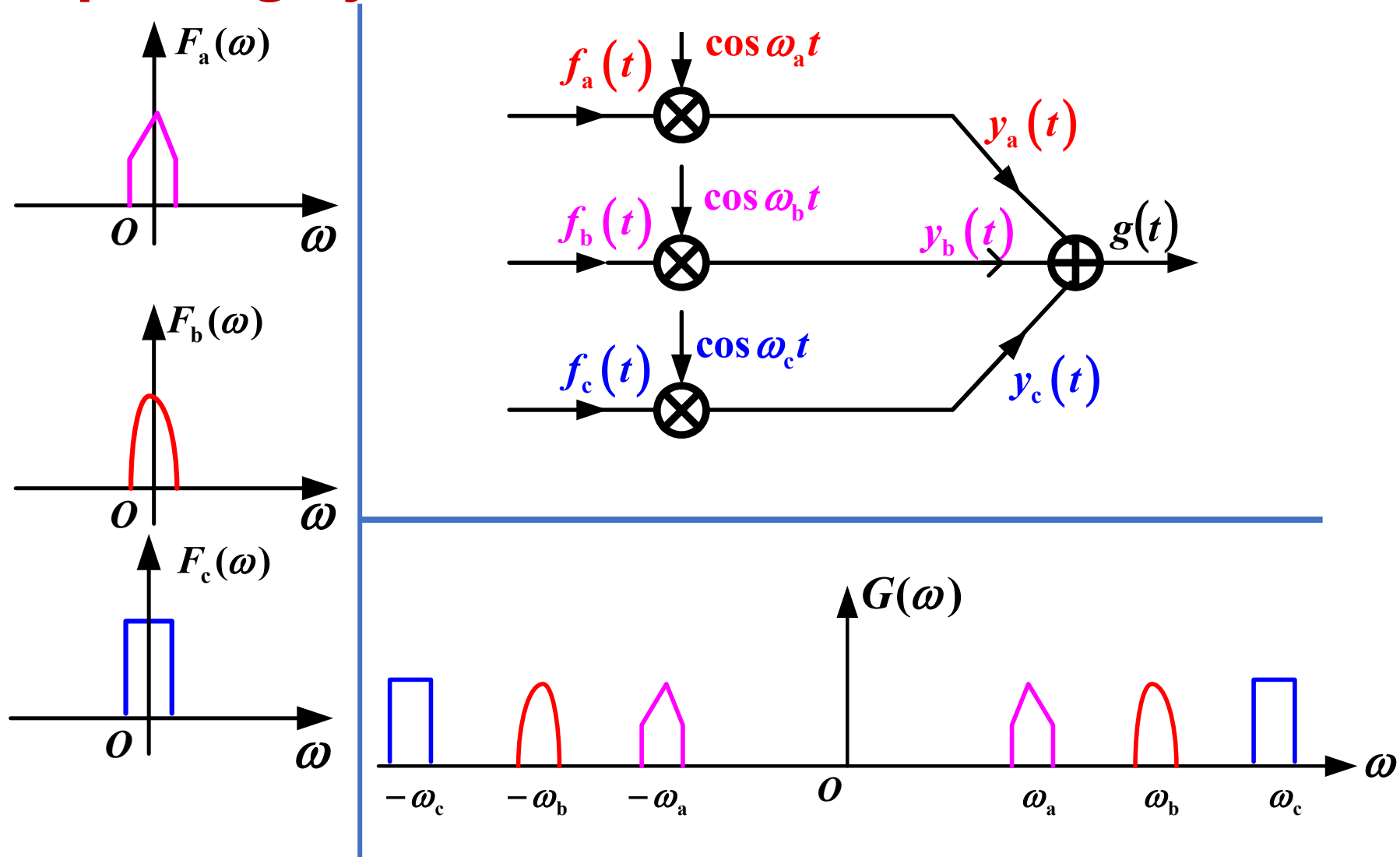
5. Frequency Division Multiplexing

(1) Definition

- Frequency Division Multiplexing (FDM)
- A transmission scheme that enables multiple communication channels within a single physical medium by dividing the available bandwidth into distinct frequency sub-bands.

3. Amplitude Modulation

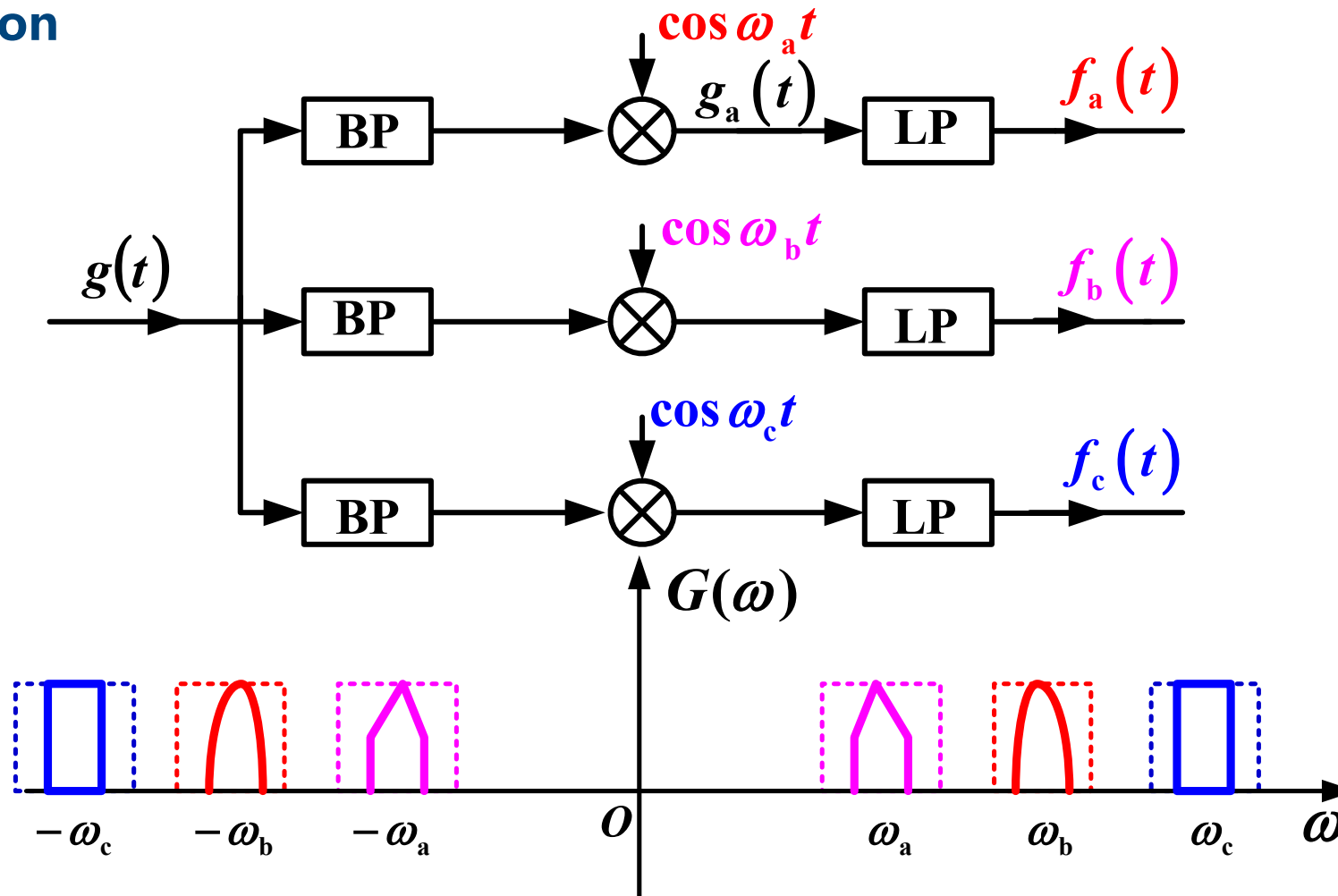
(2) Multiplexing System Structure



3. Amplitude Modulation

(3) Demultiplexing System Structure

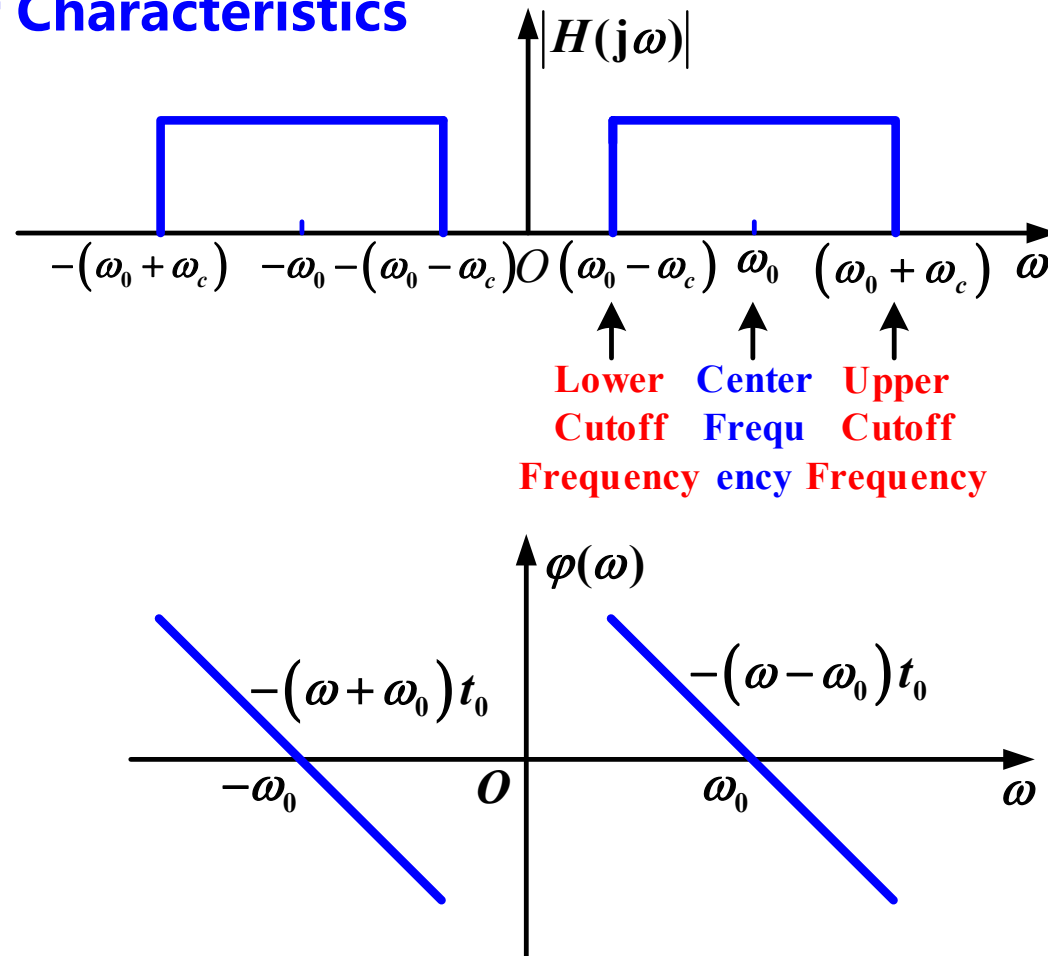
Uses **bandpass filters** to separate each channel signal, followed by individual demodulation



3. Amplitude Modulation

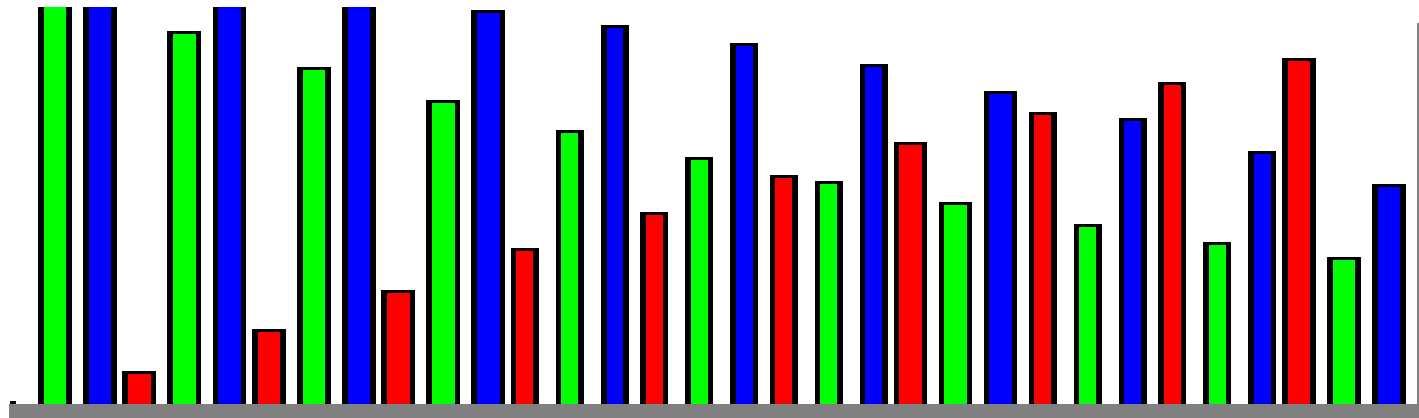
(3) Demultiplexing System Structure

Bandpass Filter Characteristics



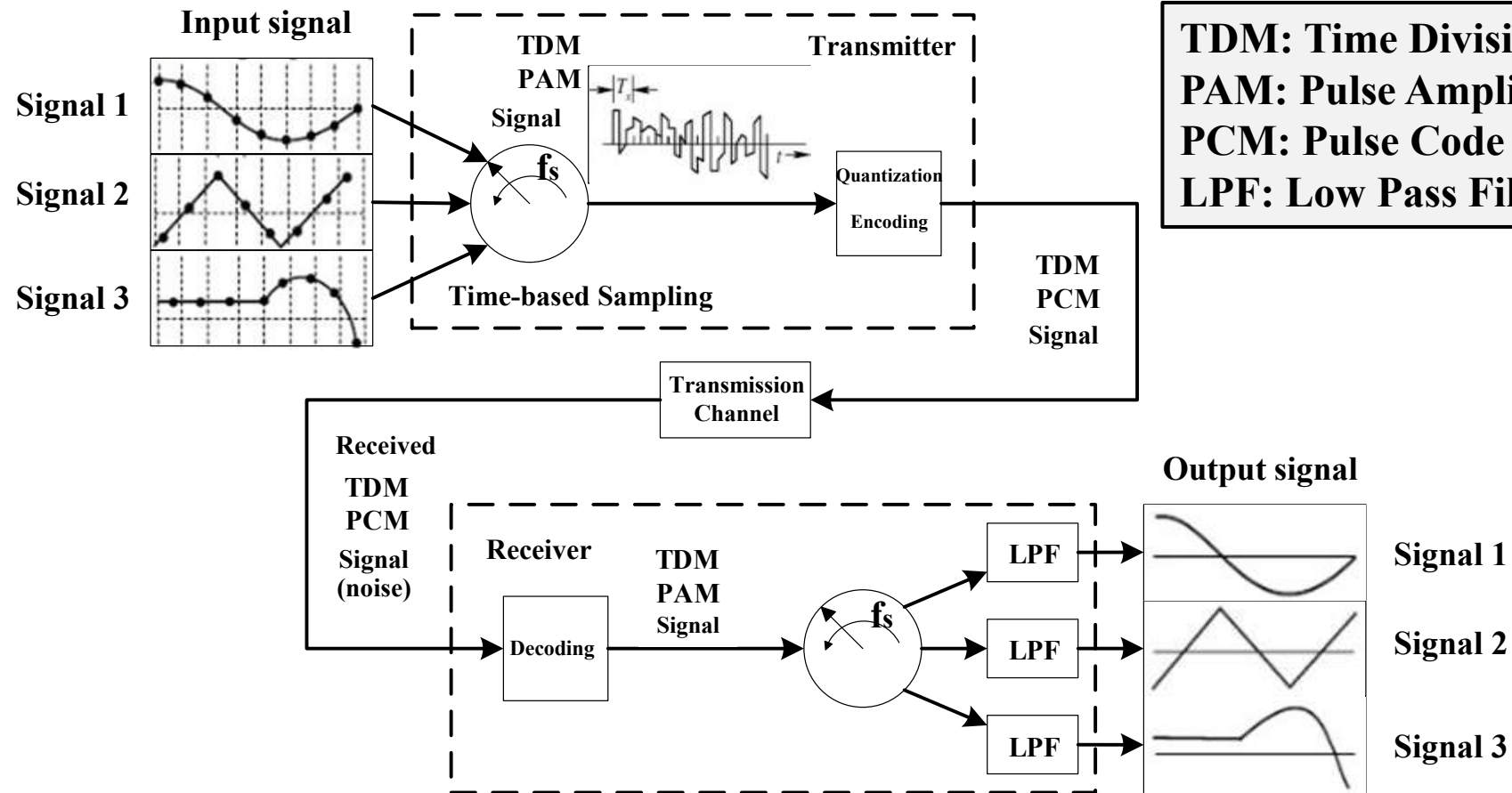
6. Time Division Multiplexing

- **Primary Application:** Digital signal transmission and access systems
- **Core Principle:** Divides the transmission channel into discrete time intervals
- **Key Terminology:** Time Slot, Each allocated segment of the transmission timeframe



3. Amplitude Modulation

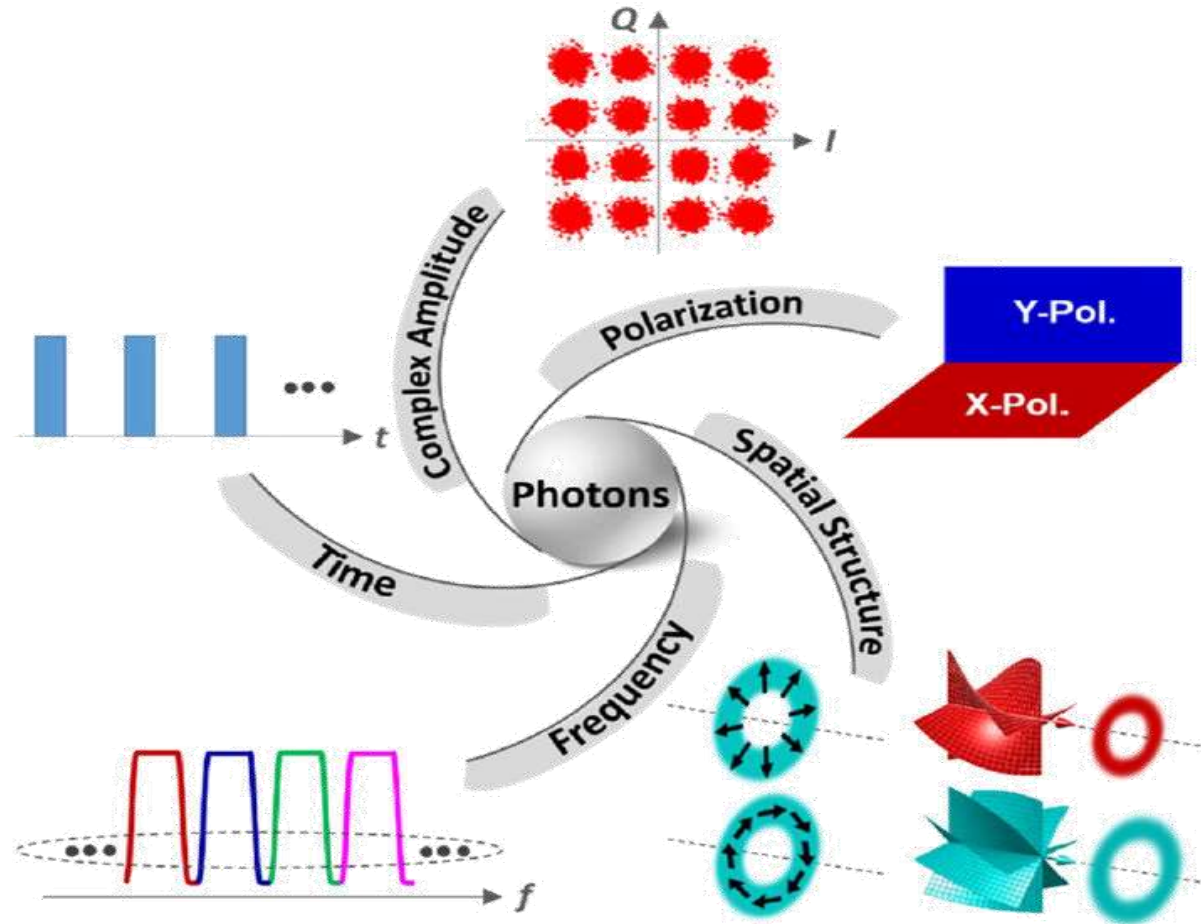
6. Time Division Multiplexing



System Diagram of TDM-PCM for Three Analog Signals

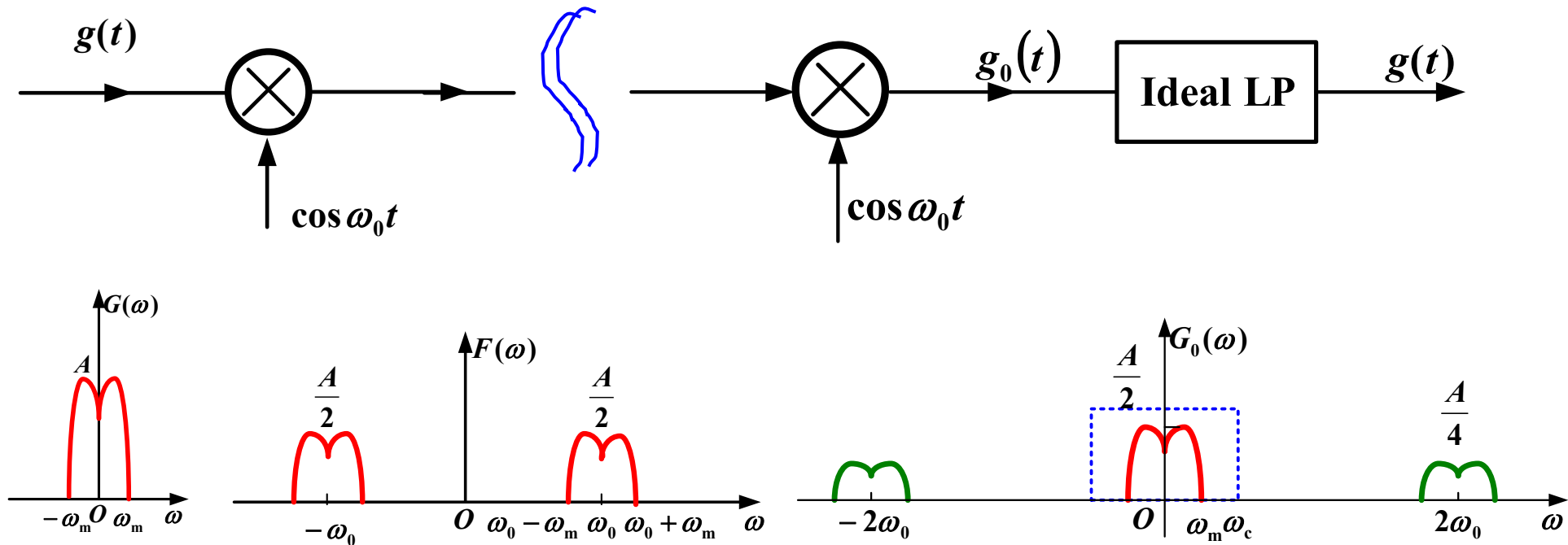
7. Brief Introduction to Other Multiplexing Methods

Multiplexing: The technique of transmitting multiple signals over a single communication channel



3. Amplitude Modulation

Summary



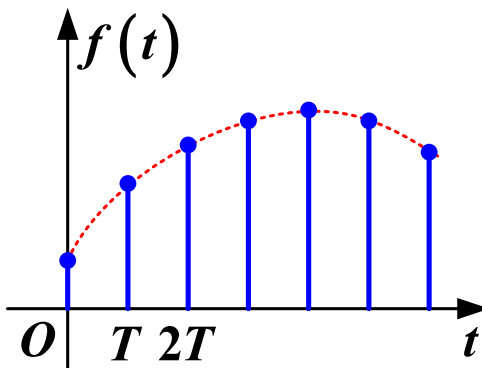
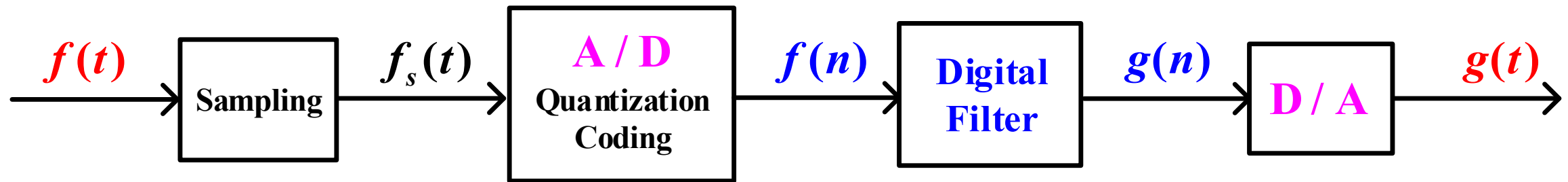
Main Content

- **Concept of Sampling**
- ✓ **Digital Processing of Analog Signals**
- ✓ **Sampling Interval and Sampling Frequency**
- **Spectrum of Sampled Signals**
- **Time-Domain Sampling Theorem**
- **Reconstruction of Continuous-Time Signals from Samples**
- ✓ **Frequency-Domain Filtering**
- ✓ **Time-Domain Interpolation**
- **Aliasing**

4. Sampling Theorem and Aliasing

1. Concept of Sampling

(1) Digital Processing of Analog Signals



$$f_d(n) = f(t)|_{t=nT}$$

T : Sampling interval

$$f_s = \frac{1}{T} : \text{Sampling frequency}$$

Problems to be addressed:

- (1) How to set the **sampling frequency**?
- (2) Can $f(t)$ be **reconstructed** from $f_s(t)$?

(2) How to Set the Sampling Frequency?

Principle of Motion Pictures:

The fundamental principle of film is "*persistence of vision*". Scientific experiments show that after an image disappears, the human eye retains its impression on the retina for approximately **0.1–0.4 seconds**. When film reels advance at **24 frames (samples) per second**, the rapid succession of static images leverages this physiological phenomenon to create a seamless illusion of motion

Key Considerations for Frame Rate Selection:

- Too high a sampling frequency: Introduces data redundancy
- Too low a sampling frequency: Causes motion distortion (e.g., judder or flicker)

4. Sampling Theorem and Aliasing

(2) How to Set the Sampling Frequency?



Sampling Rate
6kHz



Sampling Rate
12kHz



Sampling Rate
18kHz



Impact of Different Sampling Frequencies on Images

4. Sampling Theorem and Aliasing

(2) How to Set the Sampling Frequency?

Mosquito Wingbeat Frequency: ~600 Hz



High-Speed Recording, Standard Playback

Sampling Frequency/Frame Rate: Determined by the target frequency characteristics

4. Sampling Theorem and Aliasing

(2) How to Set the Sampling Frequency?

Stellar Rotation Frequency: $1/86,400$ Hz



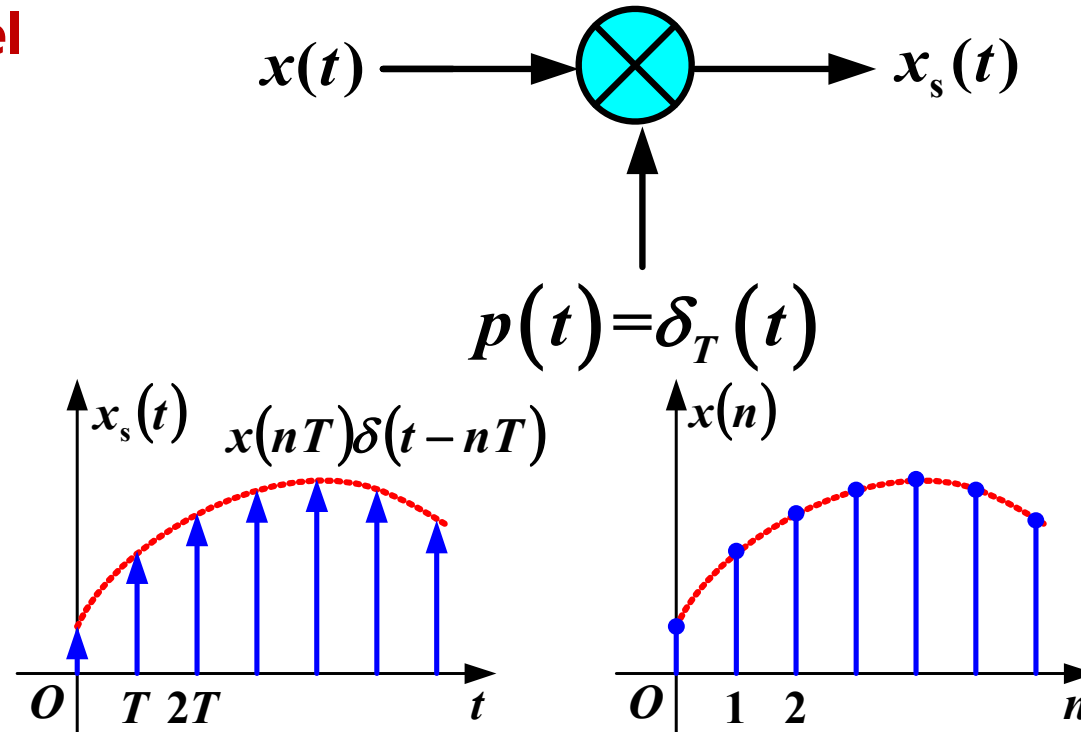
Time-Lapse Recording, Real-Time Playback

Sampling Frequency/Frame Rate: Determined by the target frequency characteristics

4. Sampling Theorem and Aliasing

2. Spectrum of Sampled Signals

(1) Sampling Model



Mathematical Model of Sampled Signals (Continuous-time to Discrete Sequence):

$$x_s(t) = x(t)\delta_T(t)$$

$$x(n) = x(t)\big|_{t=nT}$$

2. Spectrum of Sampled Signals

(2) Spectrum of the Sampled Signal

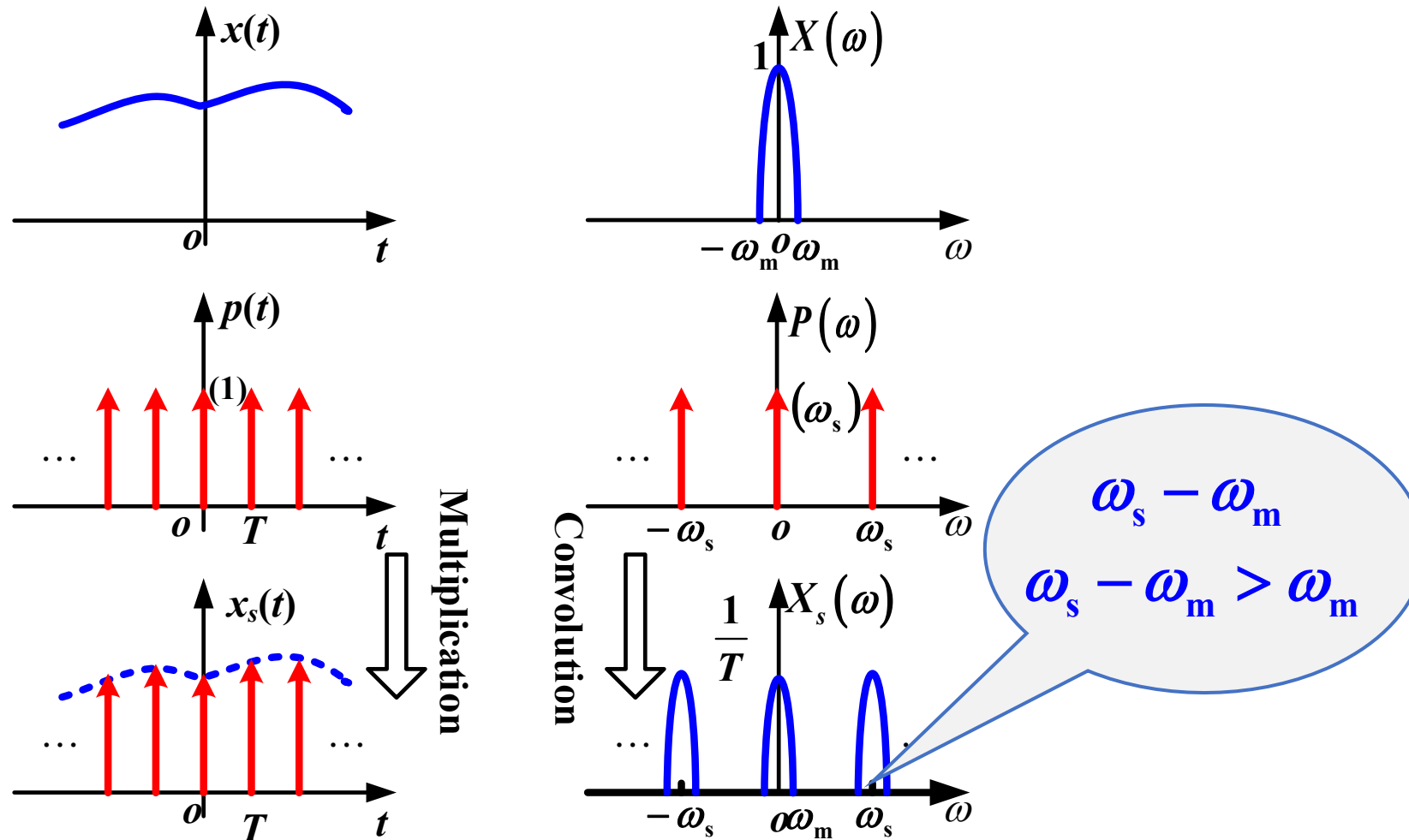
$$p(t) = \delta_T(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT)$$

$$P(\omega) = \omega_s \sum_{n=-\infty}^{\infty} \delta(\omega - n\omega_s)$$

$$\begin{aligned} X_s(\omega) &= \frac{1}{2\pi} X(\omega) * \left[\omega_s \sum_{n=-\infty}^{\infty} \delta(\omega - n\omega_s) \right] \\ &= \frac{1}{T} \sum_{n=-\infty}^{\infty} X(\omega - n\omega_s) \end{aligned}$$

4. Sampling Theorem and Aliasing

2. Spectrum of Sampled Signals



3. Sampling Theorem

- (1) For a band-limited signal $f(t)$ with a spectrum confined to $-\omega_m \sim +\omega_m$ (bandwidth: ω_m)
- (2) The signal $f(t)$ can be uniquely represented by uniformly spaced samples, provided the sampling interval T satisfies

$$T \leq \frac{1}{2f_m} \quad (\omega_m = 2\pi f_m)$$

- (3) Perfect reconstruction of $f(t)$ from $f_s(t)$ is achievable using an ideal low-pass filter with a gain of T , Cutoff frequency:

$$\omega_m < \omega_c < \omega_s - \omega_m$$

4. Sampling Theorem and Aliasing

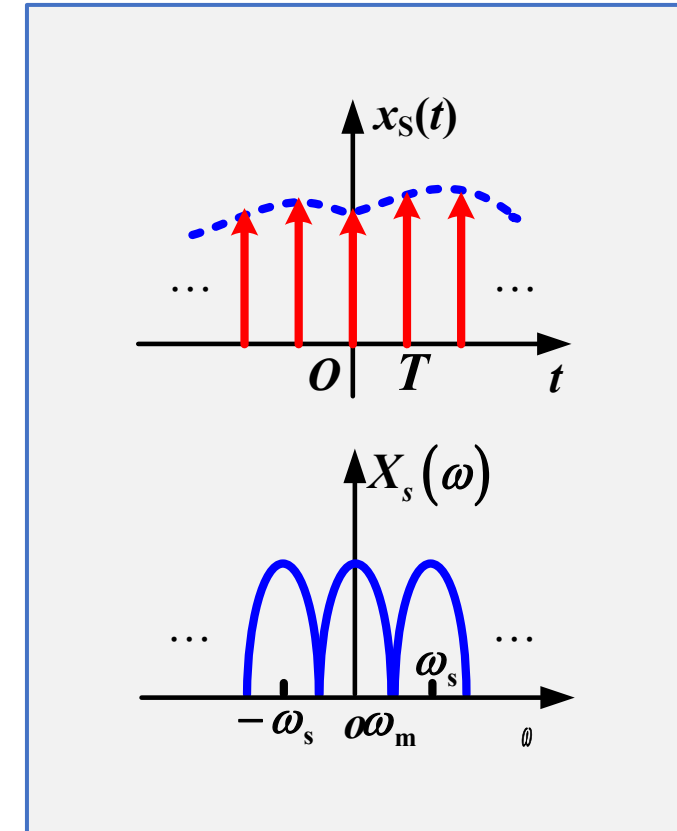
3. Sampling Theorem

$$\omega_s = 2\omega_m, f_s = 2f_m$$

Minimum allowable sampling frequency:
Known as the "Nyquist sampling frequency"

$$T = \frac{1}{2f_m}$$

Maximum allowable sampling interval:
Known as the "Nyquist sampling interval"



This nomenclature honors the pioneering contributions of **American physicist Harry Nyquist** (1889–1976) in the field of data transmission

4. Sampling Theorem and Aliasing

Example 1

For example, in audio signals, after applying an anti-aliasing filter to remove high-frequency components (spectrum range: 0–3.4 kHz), how should the sampling rate be selected?

$$f_m = 3.4 \text{ kHz}, f_s \geq 2f_m$$

$$f_{s\min} = 6800 \text{ Hz}, T_{s\max} = \frac{1}{2f_m}$$

$$\text{If let } f_s = 8000 \text{ Hz, then } T = \frac{1}{8000} = 125 \mu\text{s}$$

4. Sampling Theorem and Aliasing

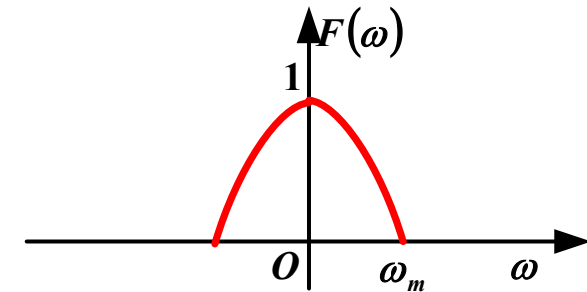
Example 2

For a band-limited signal $f(t)$ with a maximum frequency of 100 Hz, if the following signals are sampled in the time domain, determine the minimum sampling frequency $f_{\min}$

$$(1) f_1(t) = f(3t) \quad (2) f_2(t) = f(t) * f(2t)$$

$$(1) f_{1m} = 300 \text{ Hz} \quad f_{1s} \geq 600 \text{ Hz}$$

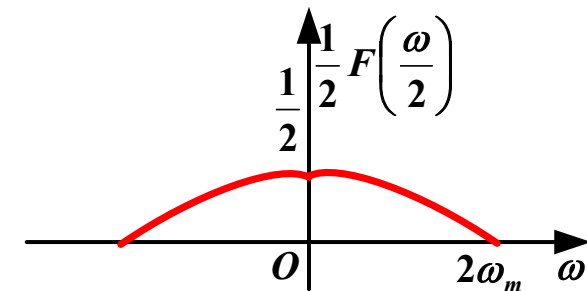
$$(2) f(t) * f(2t) \leftrightarrow F(\omega) \cdot \frac{1}{2} F\left(\frac{\omega}{2}\right)$$



$f(2t)$, bandwidth of 200 Hz

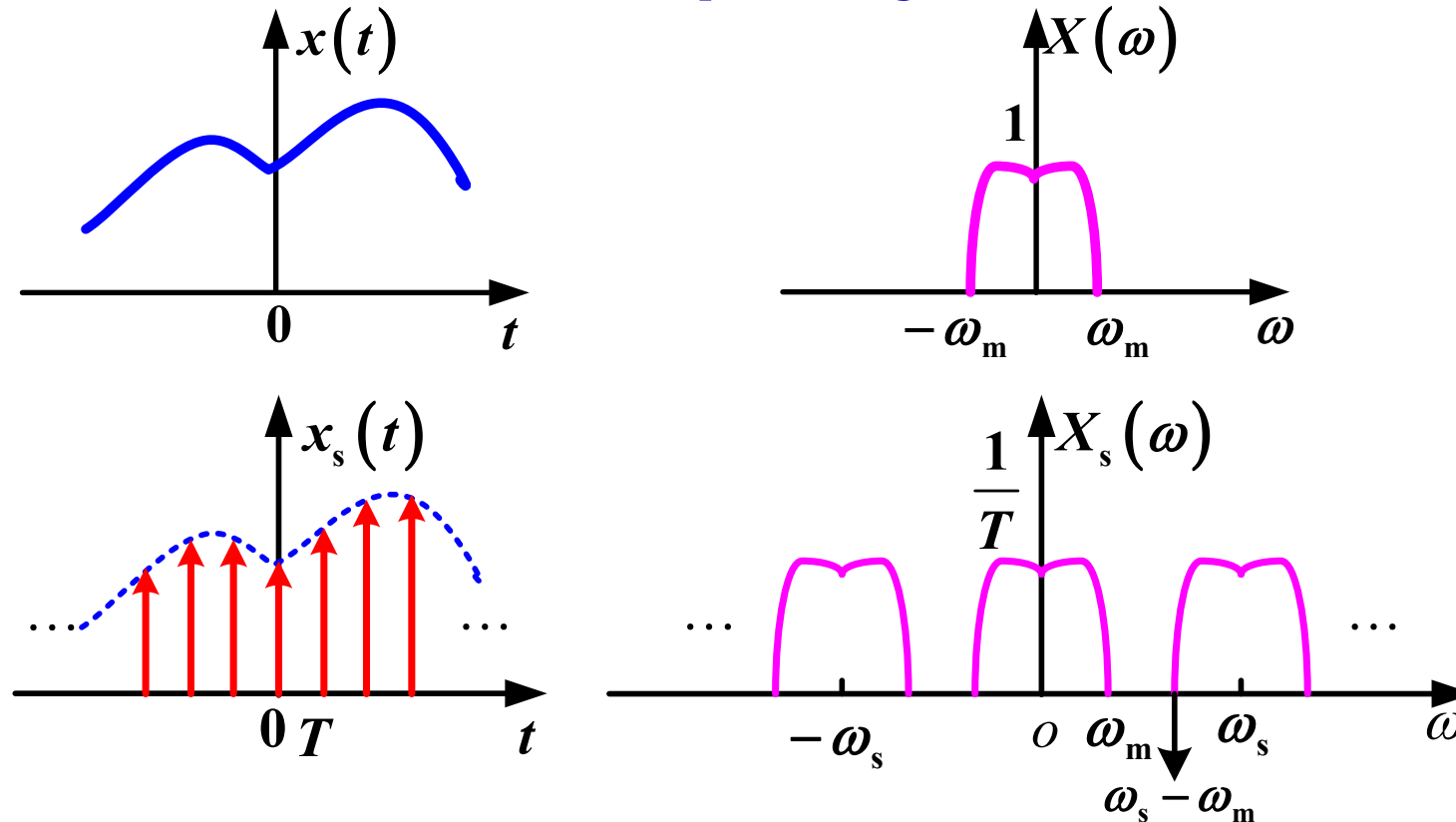
$$f_{2m} = 100 \text{ Hz}$$

$$f_{2s} \geq 200 \text{ Hz}$$



4. Sampling Theorem and Aliasing

4. Signal Reconstruction-Frequency Domain Filtering



Sampling Theorem:

- (1) Bandwidth requirement, (2) Rate requirement (2x relationship)
- (3) Recovery: Low-pass filter

4. Sampling Theorem and Aliasing

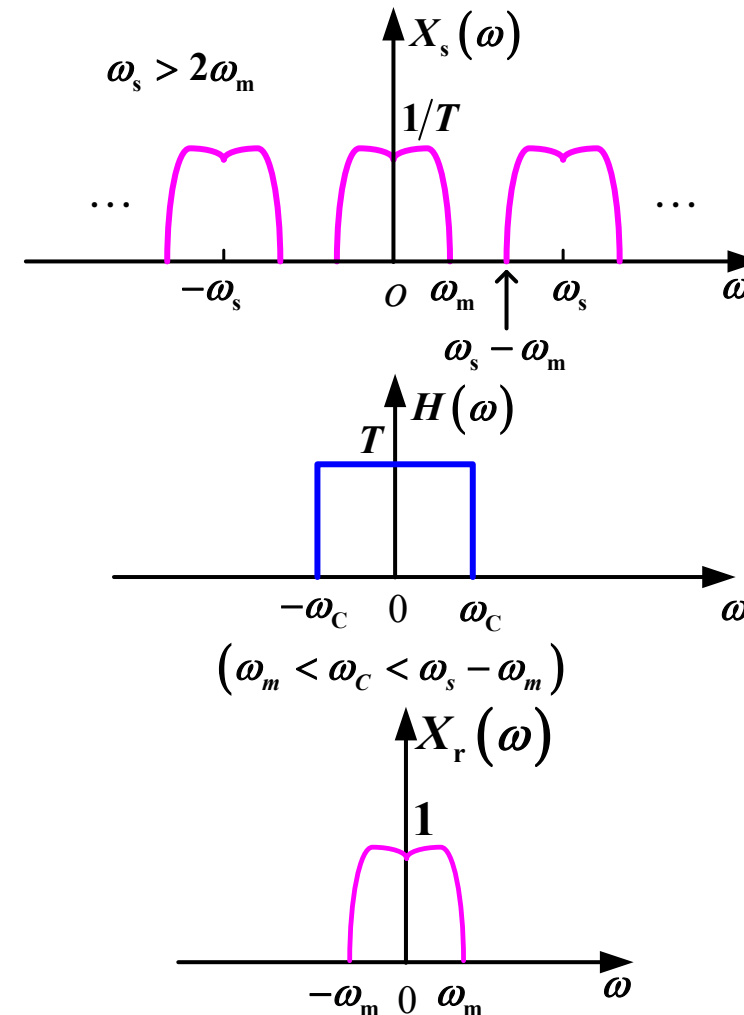
4. Signal Reconstruction-Frequency Domain Filtering

Ideal Low-Pass Filter:

$$H(\omega) = \begin{cases} T & |\omega| < \omega_c \\ 0 & |\omega| > \omega_c \end{cases}$$

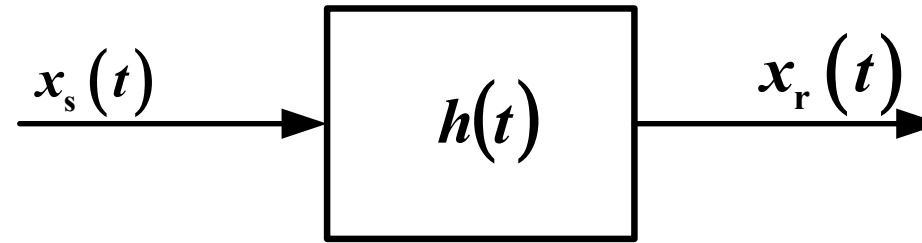
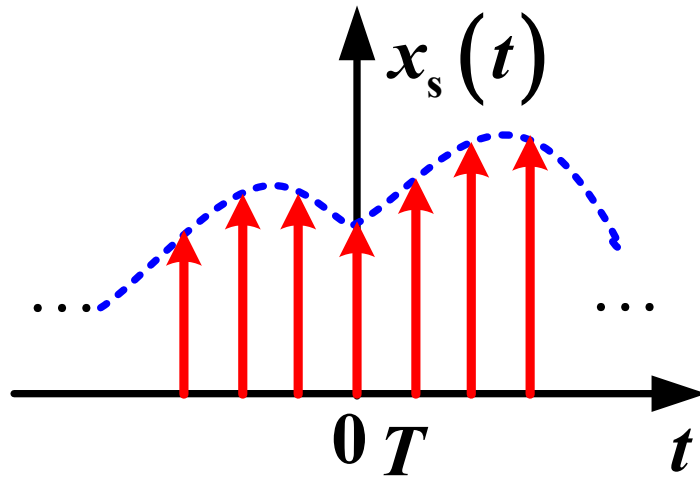
$$X_r(\omega) = X_s(\omega) \cdot H(\omega)$$

**Removing high-frequency components
reconstructs the original signal**

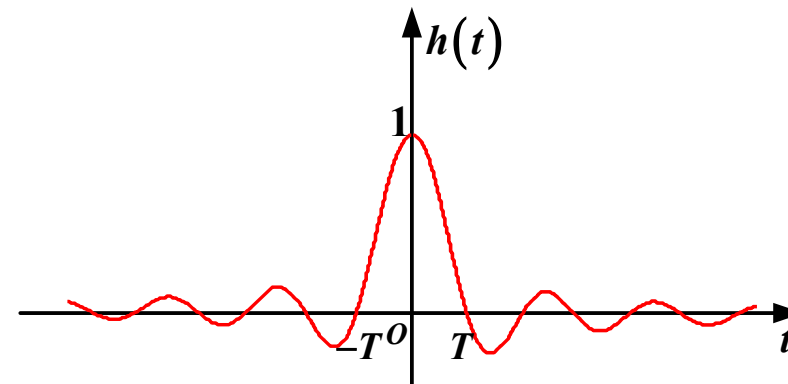


4. Sampling Theorem and Aliasing

4. Signal Reconstruction-Interpolation



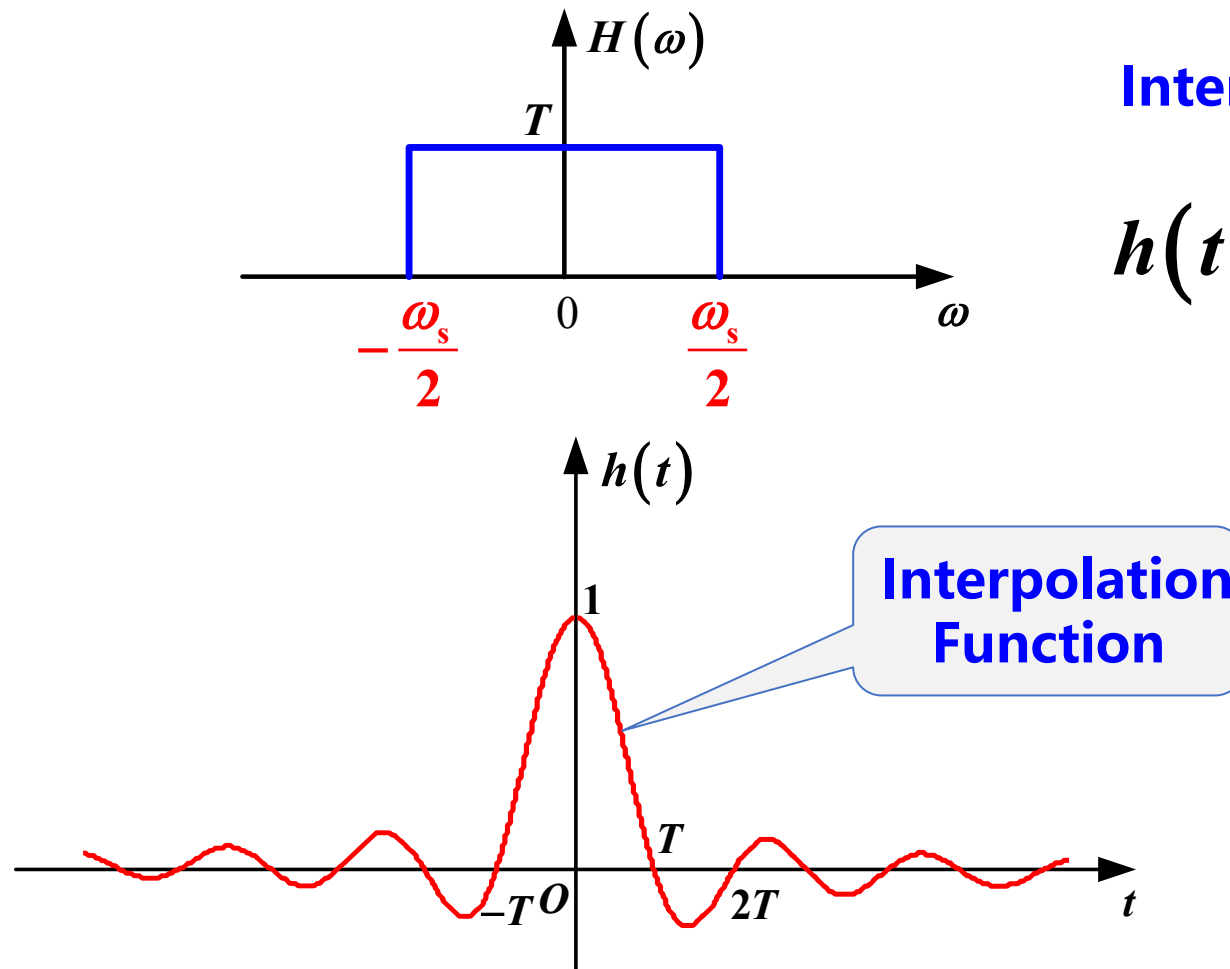
$$x_r(t) = x_s(t) * h(t)$$



Sinc function

4. Sampling Theorem and Aliasing

(1) Sa Function as Ideal Interpolation Kernel



Interpolation Function

$$h(t) = \frac{1}{2\pi} T \omega_s \text{Sa}\left(\frac{\omega_s}{2} t\right)$$
$$= \text{sinc}\left(\frac{1}{T} t\right)$$

4. Sampling Theorem and Aliasing

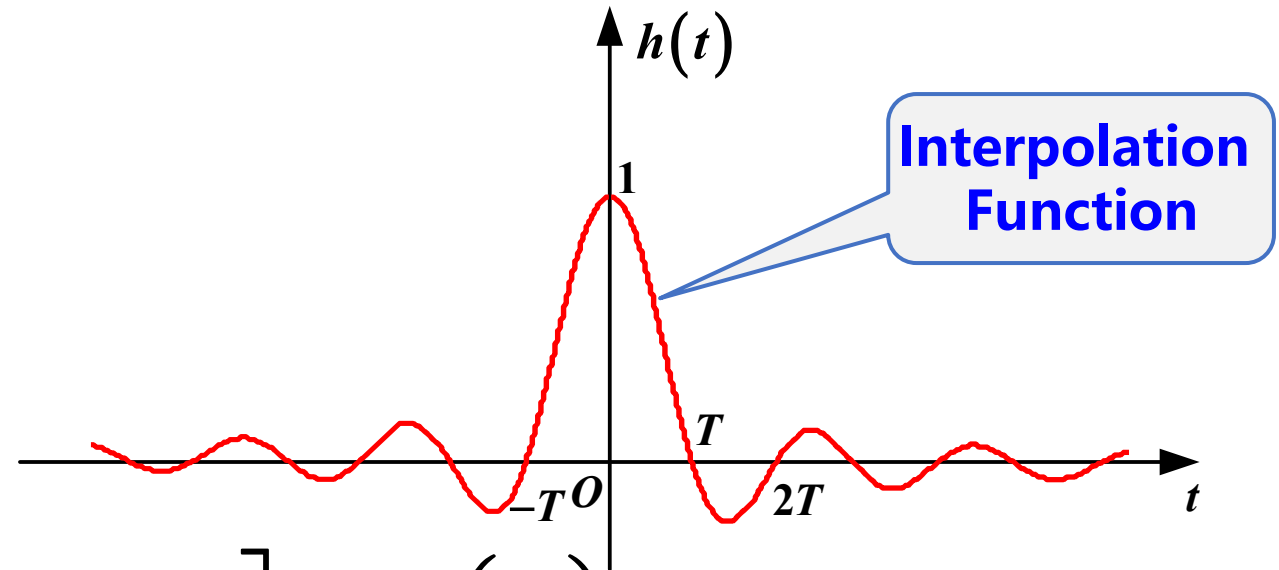
(1) Sa Function as Ideal Interpolation Kernel

$$h(t) = \text{sinc}\left(\frac{1}{T}t\right)$$

$$x_r(t) = x_s(t) * h(t)$$

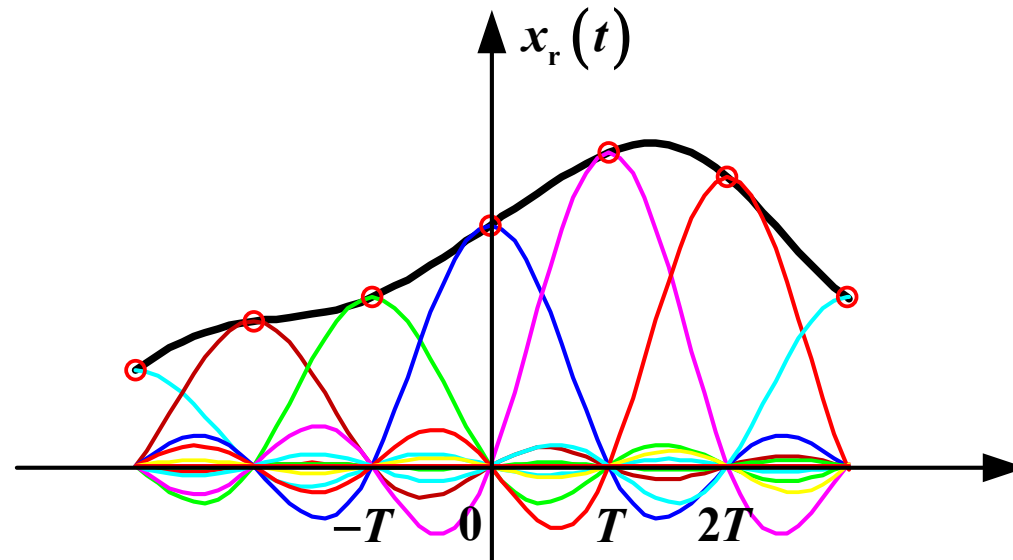
$$= \left[\sum_{n=-\infty}^{\infty} x(nT) \delta(t - nT) \right] * \text{sinc}\left(\frac{t}{T}\right)$$

$$= \sum_{n=-\infty}^{\infty} x(nT) \left[\text{sinc}\left(\frac{t - nT}{T}\right) \right]$$



4. Sampling Theorem and Aliasing

(1) Sa Function as Ideal Interpolation Kernel



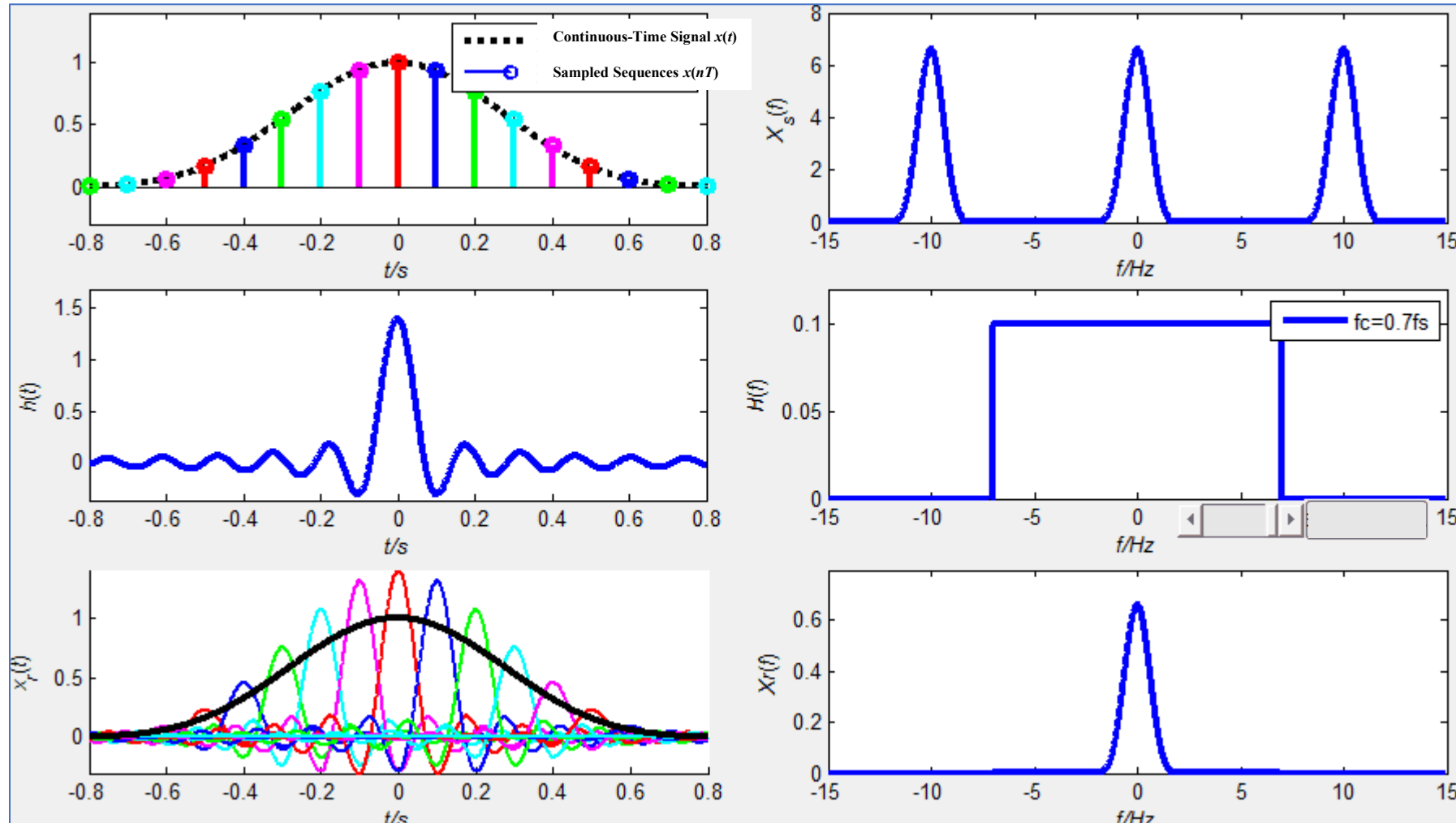
- The **zeros** of the sinc interpolation function are spaced exactly at the **sampling interval T**
- **At any sampling point**, the sum of their values **equals the sample value $x(nT)$** at that point
- **At non-sampling points**, the reconstructed signal equals the superposition of multiple sinc functions, enabling **perfect recovery** of the original continuous-time signal

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$$\omega_c = \frac{\omega_s}{2}$$

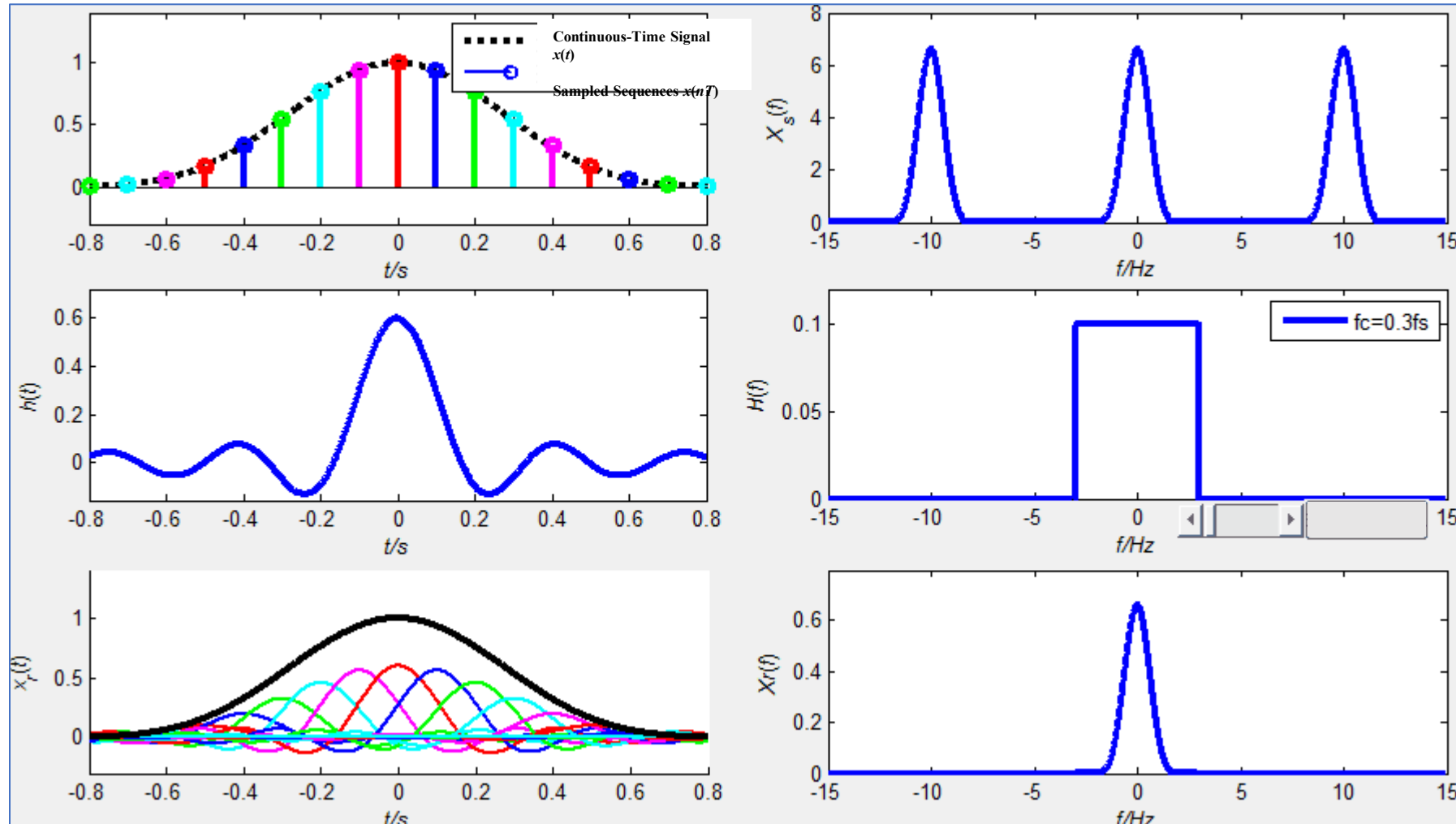

4. Sampling Theorem and Aliasing

(1) Sa Function as Ideal Interpolation Kernel $\frac{\omega_s}{2} < \omega_c < \omega_s - \omega_m$



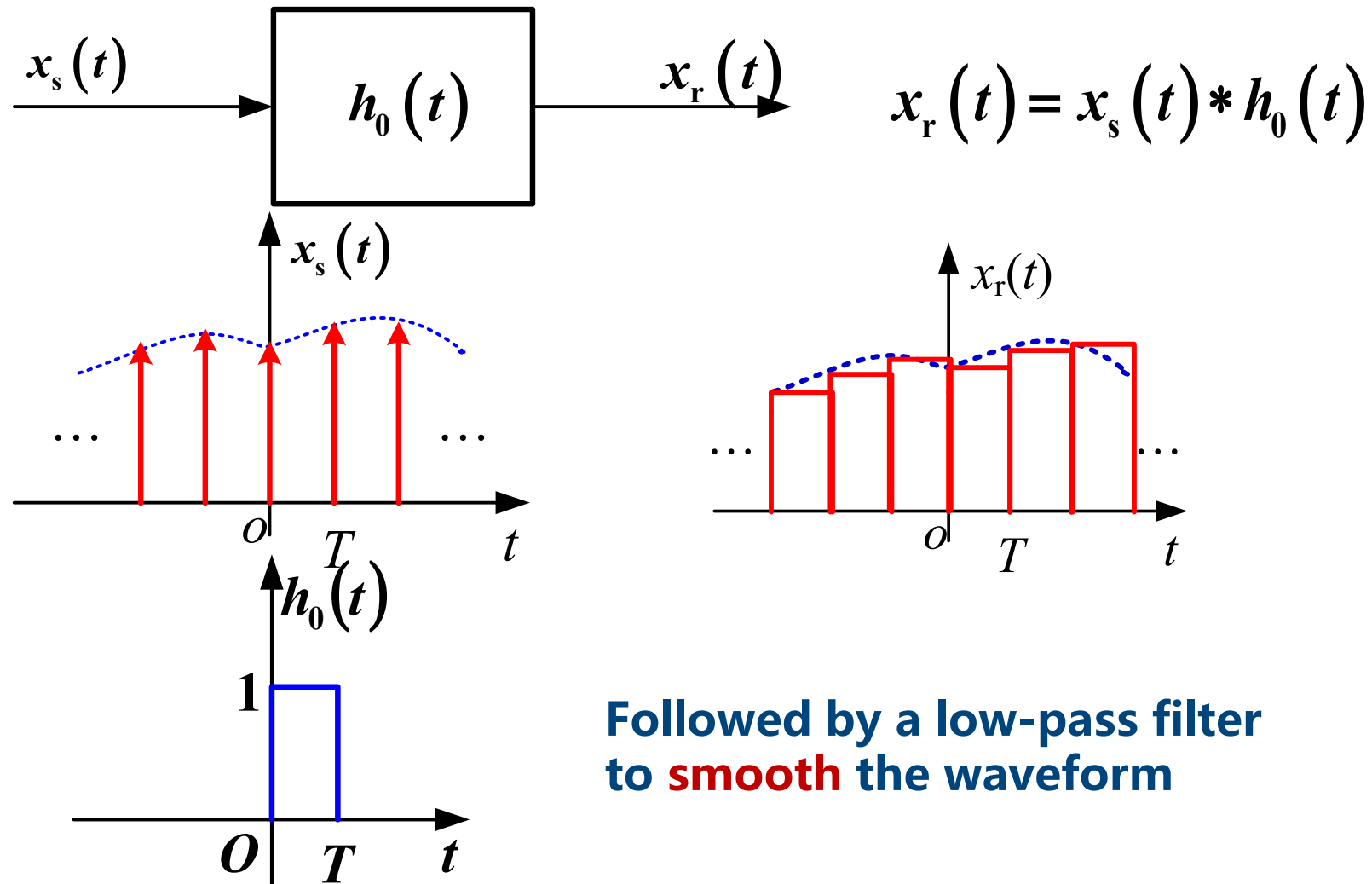
4. Sampling Theorem and Aliasing

(1) Sa Function as Ideal Interpolation Kernel $\omega_m < \omega_c < \frac{\omega_s}{2}$



4. Sampling Theorem and Aliasing

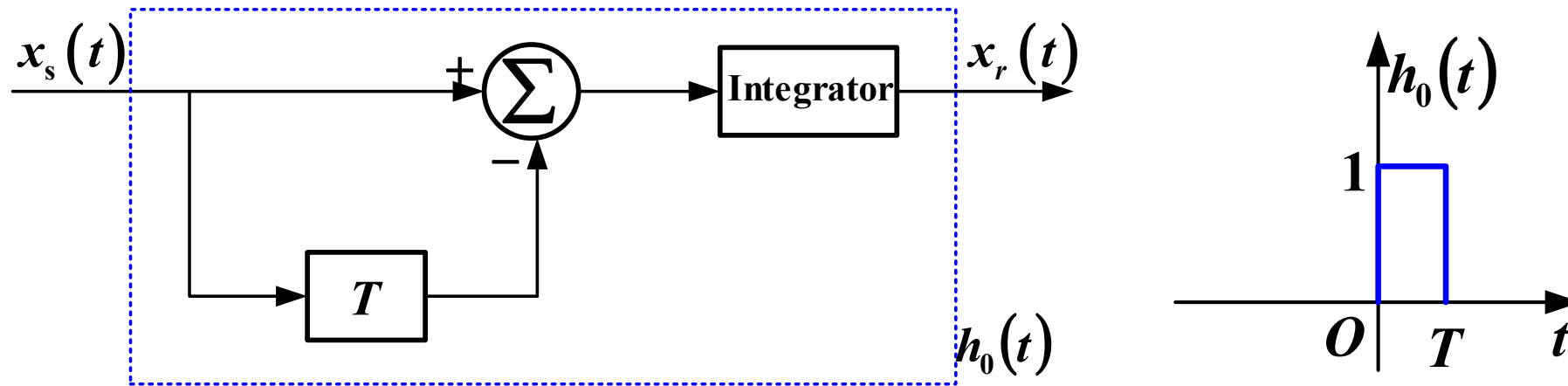
(2) Zero-order Hold (rectangular): engineering applications



4. Sampling Theorem and Aliasing

(2) Zero-order Hold (rectangular): engineering applications

Zero-Order Hold System Architecture

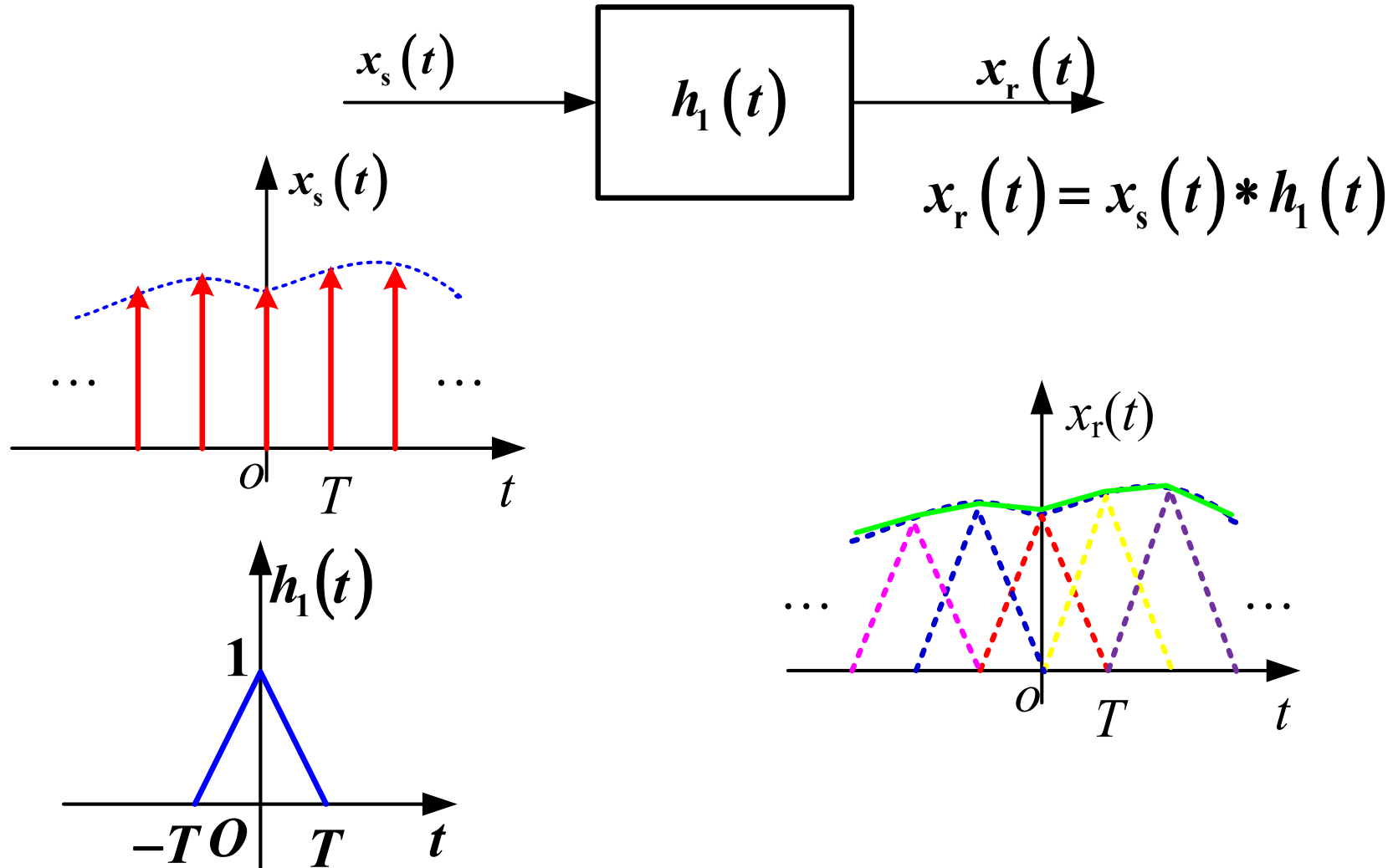


Unit Impulse Response

$$h_0(t) = \int_{-\infty}^t [\delta(t) - \delta(t - T_s)] dt = u(t) - u(t - T)$$

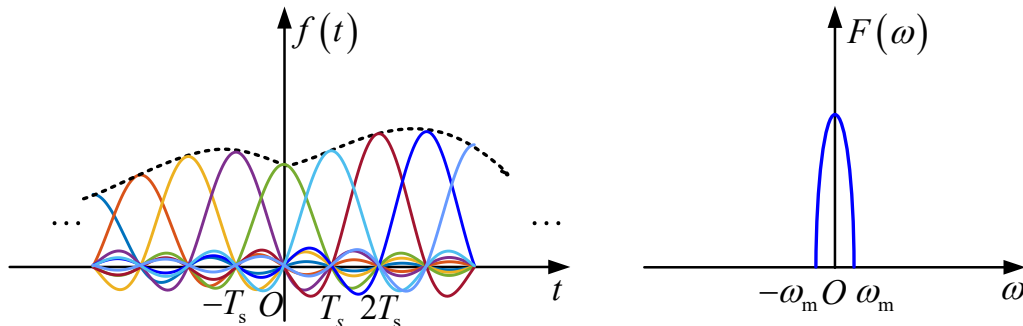
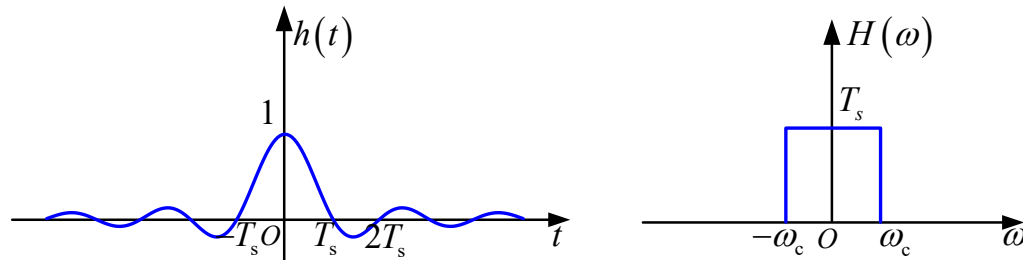
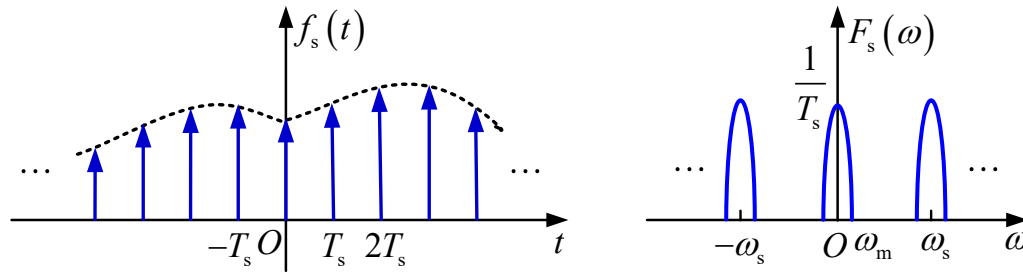
4. Sampling Theorem and Aliasing

(3) First-Order Hold (triangular)



4. Sampling Theorem and Aliasing

Summary



$$f_d(n) = f(t)|_{t=nT}$$

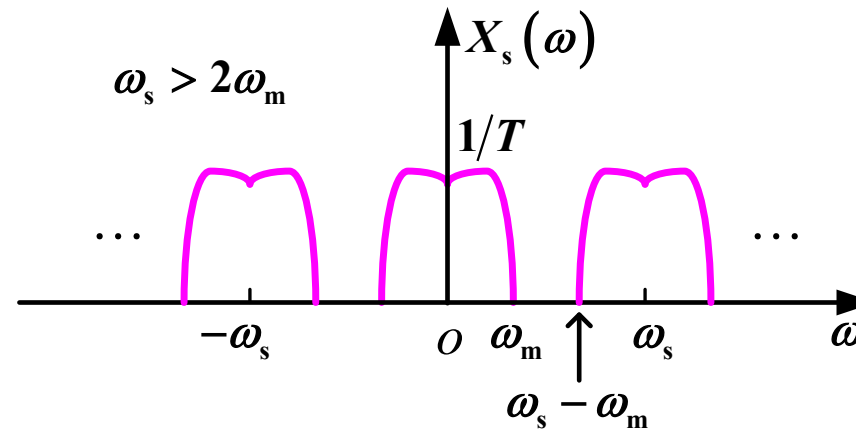
T : Sampling interval

$$f_s = \frac{1}{T} : \text{Sampling frequency}$$

Sampling Theorem: $f_s \geq 2f_m$

4. Sampling Theorem and Aliasing

5. Aliasing



$$X_s(\omega) = \frac{1}{T} \sum_{n=-\infty}^{\infty} X(\omega - n\omega_s)$$

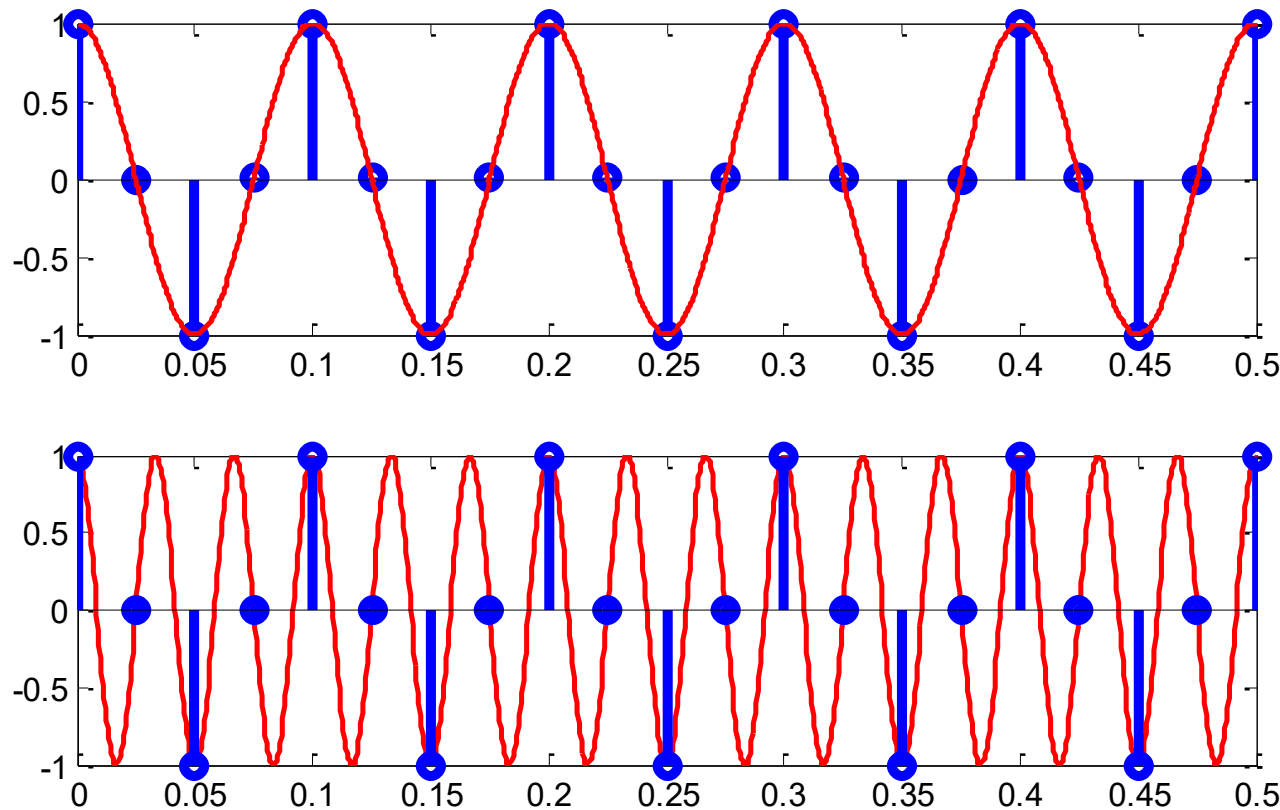
For a frequency component in real signals ω_1

Its bilateral spectrum representation consists of $\begin{cases} \omega_1 \\ -\omega_1 \end{cases}$ After Sampling: $\begin{cases} \omega_1 + n\omega_s \\ -\omega_1 + n\omega_s \end{cases}$

4. Sampling Theorem and Aliasing

Example 1

Sampling frequency is 40 Hz, sampling signals with frequencies of 10 Hz and 30 Hz. The waveforms of the sampled signals are as follows:



4. Sampling Theorem and Aliasing

Example 1

$$f_s = 40 \text{ Hz} \quad T_s = \frac{1}{f_s} = \frac{1}{40} \text{ s}$$

Continuous signal

$$x_{a1}(t) = \cos(2\pi \times 10t)$$

$$x_1(n) = x_{a1}(t)|_{t=nT_s}$$

$$= \cos\left(2\pi \times \frac{10}{40}n\right)$$

Sampled sequence

$$= \cos\left(\frac{\pi}{2}n\right)$$

$$x_{a2}(t) = \cos(2\pi \times 30t)$$

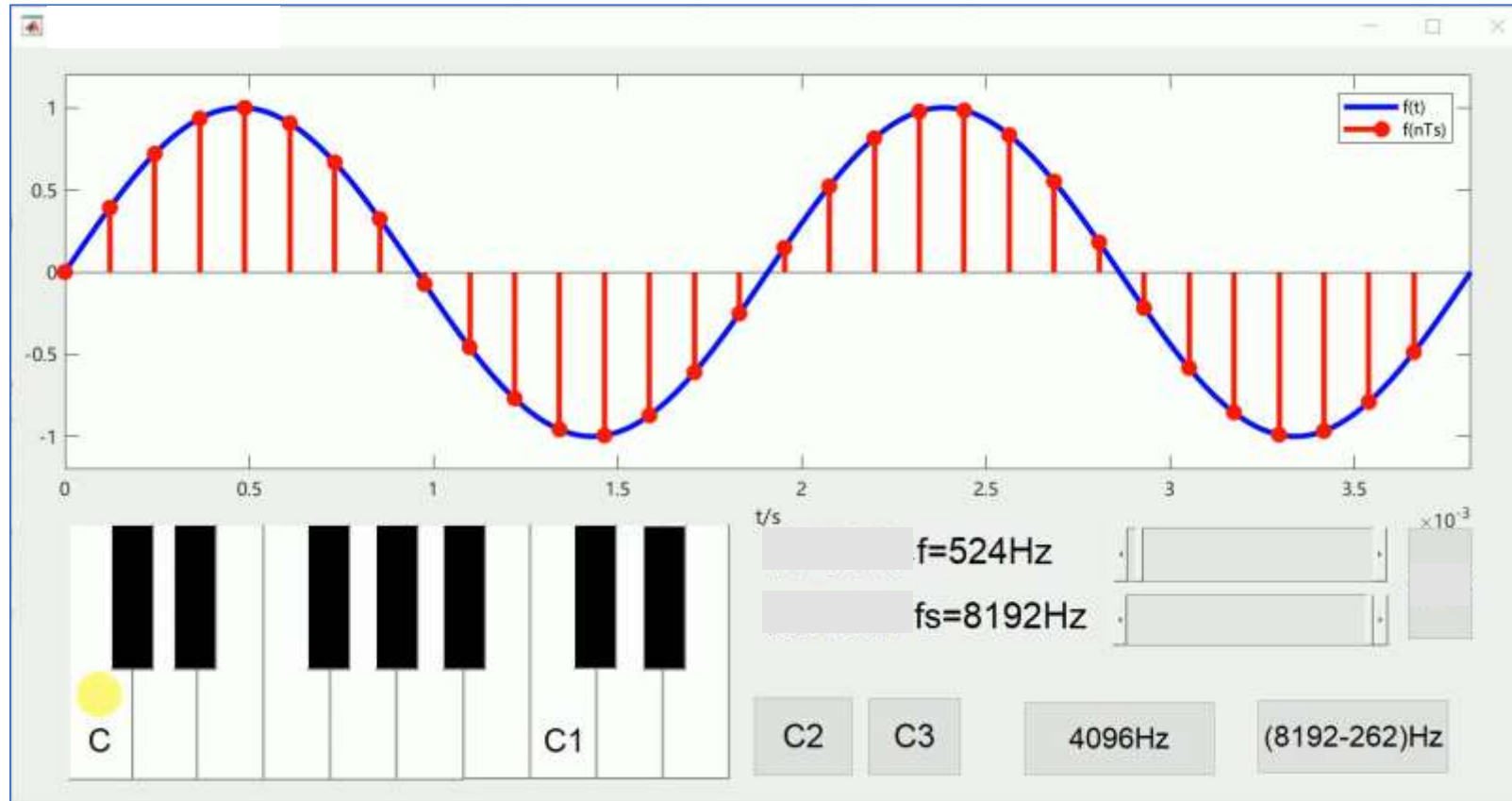
$$x_2(n) = x_{a2}(t)|_{t=nT_s}$$

$$= \cos\left[2\pi \times \frac{30}{40}n\right] = \cos\left(2\pi n - \frac{\pi}{2}n\right)$$

$$= \cos\left(\frac{\pi}{2}n\right)$$

4. Sampling Theorem and Aliasing

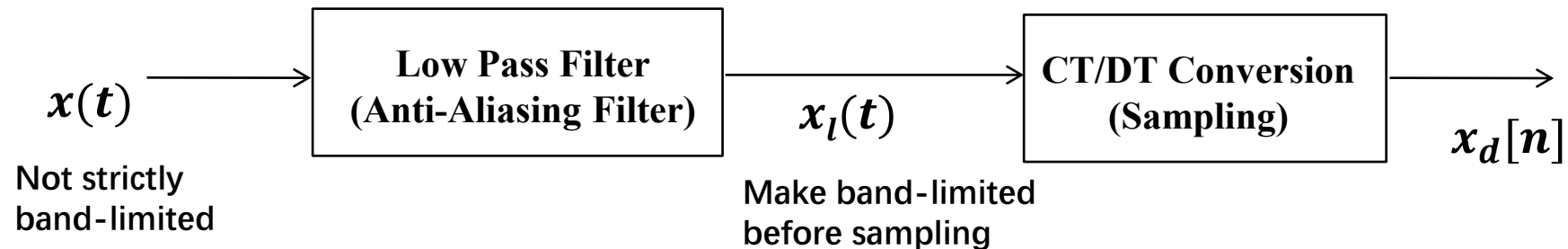
Example 1



5. Aliasing (Anti-Aliasing)

In many CT to DT conversion systems, we want to keep the sampling rate low, but the input CT signal may not be strictly band-limited to $\frac{1}{2}$ of the sampling rate

- A *anti-aliasing filter* (AAF), which is simply a low-pass filter, is often used to remove the high frequency components in the CT signal before sampling

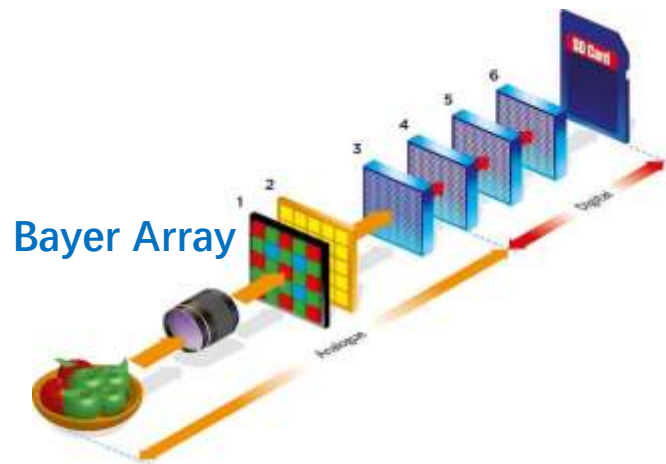


- With anti-aliasing filter, the error is *only* in the high frequency components lost
- Without anti-aliasing filter, *the high frequency components are not only lost, they turn into low frequency signals that should not be there!*

4. Sampling Theorem and Aliasing

5. Aliasing (Anti-Aliasing)

- The digital telephone network samples voice signals at 8 kHz. Therefore, filter is applied to first remove voice signal components above 4 kHz before sampling
- When we create the thumbnail of a picture, we must first low pass filter it to avoid the Moire effect
- Behind the lens of your digital camera is a filter that blurs the optical signal before spatial sampling by the CMOS sensor

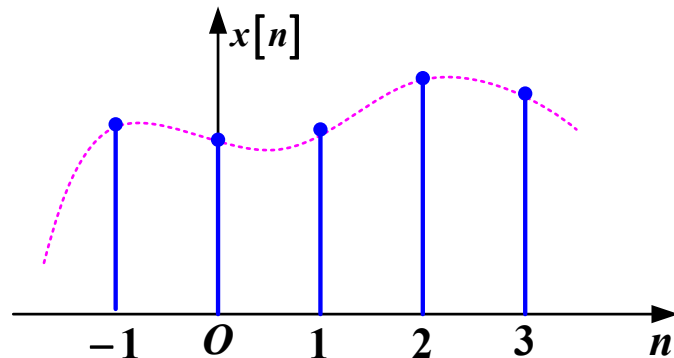
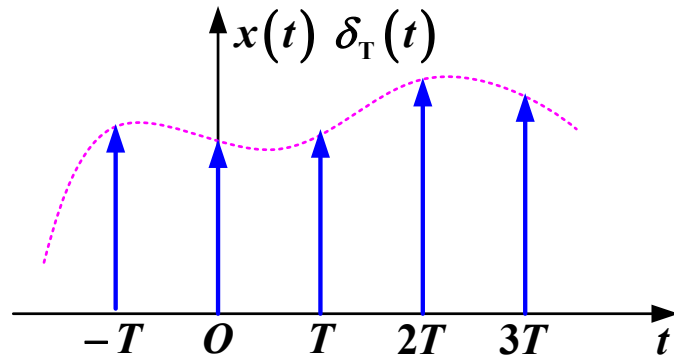


Main Content

- The Fourier Transform of a discrete-time signal (DTFT) is the Fourier transform of the sampled signal

5. Fourier Transform of a Sequence

1. Definition of DTFT



A sampled signal can be expressed as

$$\begin{aligned}x_s(t) &= x(t) \delta_T(t) \\ &= \sum_{n=-\infty}^{+\infty} x(nT) \delta(t - nT)\end{aligned}$$

Its Fourier transform

$$\begin{aligned}\mathcal{F} \left[\sum_{n=-\infty}^{\infty} x(nT) \delta(t - nT) \right] \\ = \sum_{n=-\infty}^{\infty} x(nT) e^{-jn\omega T}\end{aligned}$$

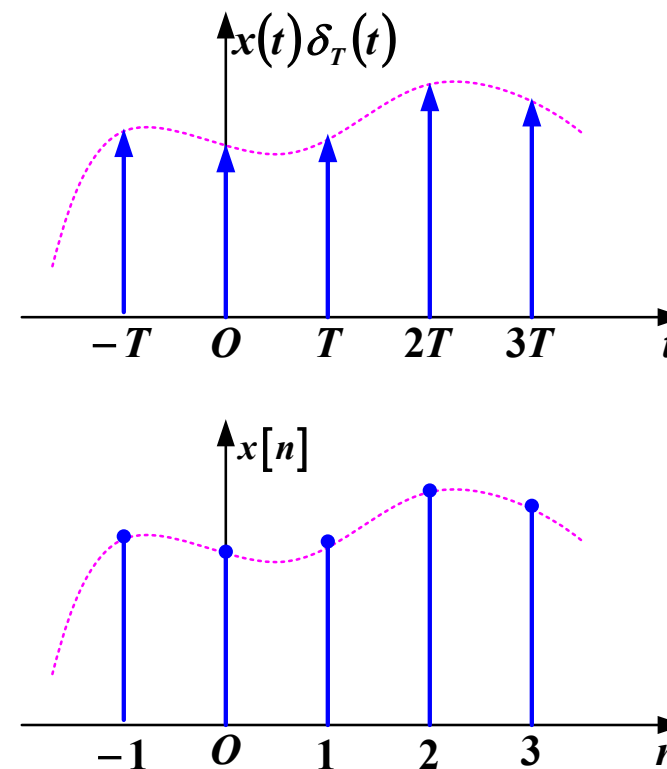
ω Analog angular frequency

5. Fourier Transform of a Sequence

1. Definition of DTFT

Let $x(nT)$ be denoted as $x[n]$, and $\omega T = \Omega$ Ω digital angular frequency

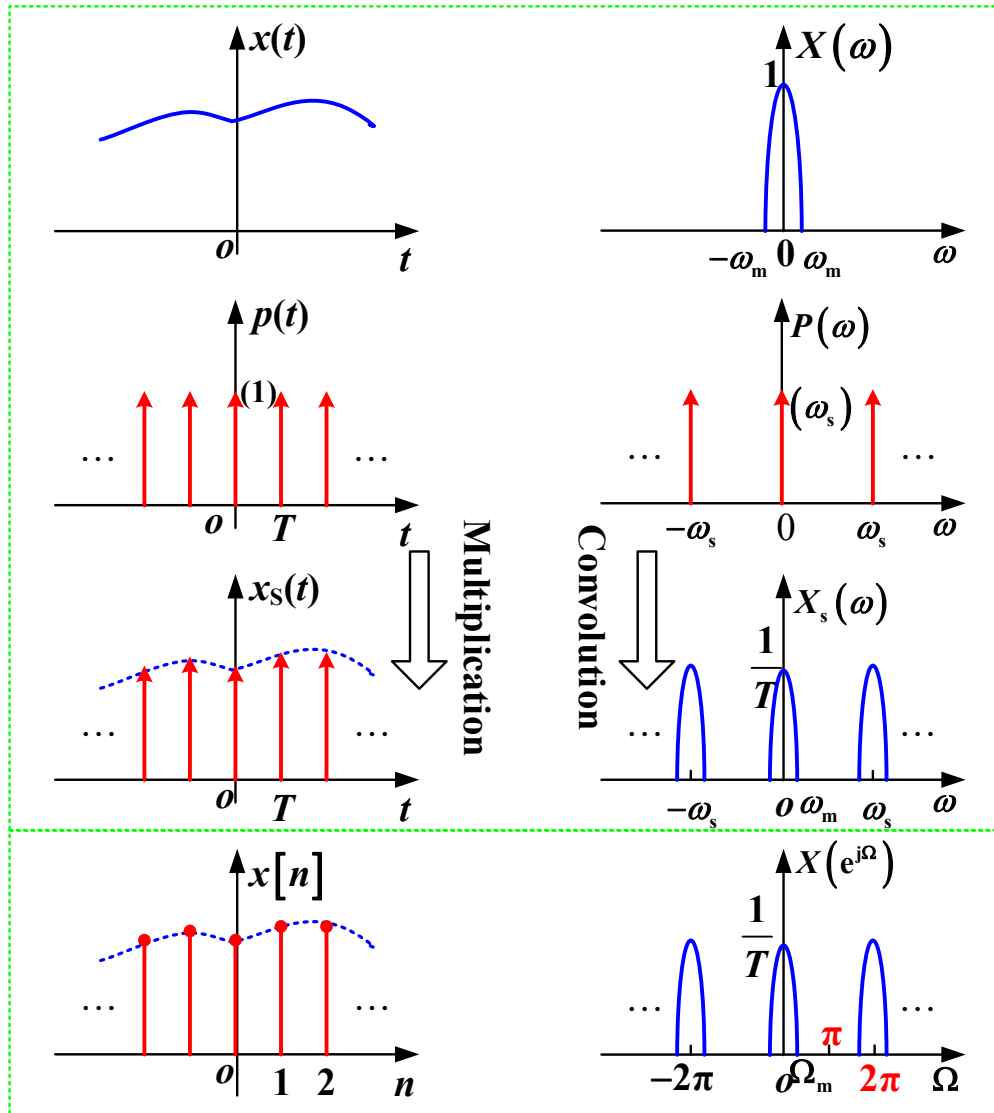
$$\begin{aligned}\mathcal{F}[x(t)\delta_T(t)] &= \sum_{n=-\infty}^{\infty} x(nT) e^{-jn\omega T} \\ &= \sum_{n=-\infty}^{\infty} x[n] e^{-jn\Omega} \\ &= \text{DTFT}[x[n]] \\ &= X(e^{j\Omega})\end{aligned}$$



This is called the Fourier Transform of a discrete-time signal
Discrete-time Fourier transform (DTFT)

5. Fourier Transform of a Sequence

1. Definition of DTFT



- DTFT is the Fourier Transform of a Sampled Signal
- DTFT has a period of 2π

$$\Omega = \omega T$$

$$\text{when } \omega = \omega_s$$

$$\Omega = \omega_s T = 2\pi$$

5. Fourier Transform of a Sequence

1. Definition of DTFT

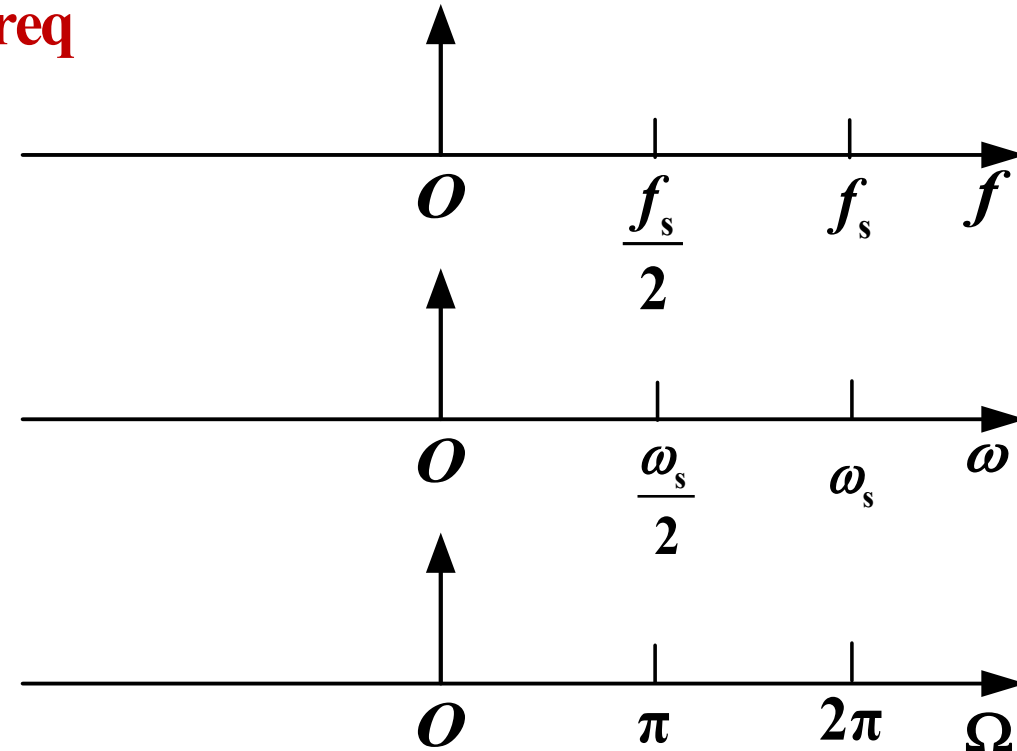
Frequency, Analog Ang Freq, and Dig Ang Freq

Frequency f

Analog Angular Frequency $\omega = 2\pi f$

Digital Angular Frequency

$$\Omega = \omega T = \frac{\omega}{f_s} = 2\pi \frac{f}{f_s}$$



Given a sampling frequency f_s , the **highest** frequency of the signal being processed is $\frac{f_s}{2}$:

$$\Omega_m \leq \pi$$

This means π is the maximum value of the digital angular frequency

1. Definition of DTFT

□ Using the frequency-domain convolution theorem

$$X_s(\omega) = \frac{1}{T_s} \sum_{n=-\infty}^{+\infty} X(\omega - n\omega_s)$$

This indicates that the spectrum of the sampled signal is a periodic extension of the original signal's spectrum

□ Using the time-shift property (DTFT)

$$X(e^{j\Omega}) = \sum_{n=-\infty}^{+\infty} x[n]e^{-jn\Omega}$$

This provides a method that can be used in engineering to compute the frequency spectrum

Main Content

- Derivation of the Discrete Time Fourier Series (DTFS)
- Properties of DTFS

1. Derivation of the Discrete Time Fourier Series (DTFS)

(1) Continuous Fourier Series Recap

A continuous periodic signal $\tilde{x}(t)$ with period T can be expressed as:

$$\tilde{x}(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$$

where:

$\omega_0 = \frac{2\pi}{T}$ (fundamental frequency),

a_k are the Fourier coefficients:

$$a_k = \frac{1}{T} \int_0^T \tilde{x}(t) e^{-jk\omega_0 t} dt$$

(2) Moving to Discrete-Time Signals

For a discrete-time periodic signal $\tilde{x}[n]$ with period N , we sample $\tilde{x}(t)$ at intervals $t = nT_s$, where T_s is the sampling period

The discrete-time signal is periodic:

$$\tilde{x}[n] = \tilde{x}[n + mN] \text{ for any integer } m$$

The fundamental digital frequency is $\Omega_0 = \frac{2\pi}{N}$

(3) Discrete-Time Fourier Series Representation

We seek to represent $\tilde{x}[n]$ as a sum of complex exponentials:

$$\tilde{x}[n] = \sum_{k=0}^{N-1} \tilde{X}[k] e^{jk\Omega_0 n}$$

where:

$\tilde{X}[k]$ are the DFS coefficients(discrete frequency components),

The summation is over N terms because the signal is periodic in N

(4) Orthogonality of Discrete Exponentials

The key to finding $\tilde{X}[k]$ is the orthogonality of discrete complex exponentials:

$$\sum_{n=0}^{N-1} e^{j\frac{2\pi}{N}(k-r)n} = \begin{cases} N & \text{if } k = r \\ 0 & \text{otherwise.} \end{cases}$$

This means:

If $k \neq r$, the sum is zero.

Exponentials are distinct and orthogonal; Analogous to how sine/cosine waves of different frequencies integrate to zero over a period

If $k = r$, the sum is N .

(5) Solving for the DFS Coefficients $X[k]$

Multiply both sides of the DFS representation by $e^{-jr\Omega_0 n}$ and sum over one period N :

$$\sum_{n=0}^{N-1} \tilde{x}[n] e^{-jr\Omega_0 n} = \sum_{n=0}^{N-1} \left(\sum_{k=0}^{N-1} \tilde{X}[k] e^{jk\Omega_0 n} \right) e^{-jr\Omega_0 n}$$

Using orthogonality, the right-hand side simplifies to:

$$\sum_{k=0}^{N-1} \tilde{X}[k] \sum_{n=0}^{N-1} e^{j(k-r)\Omega_0 n} = N \tilde{X}[r] \quad k = r$$

Thus, the DFS analysis equation is:

$$\tilde{X}[k] = \frac{1}{N} \sum_{n=0}^{N-1} \tilde{x}[n] e^{-j\frac{2\pi}{N}kn}$$

(6) Summary of DFS Equations

Synthesis (Inverse DFS):

$$\tilde{x}[n] = \sum_{k=0}^{N-1} \tilde{X}[k] e^{j\frac{2\pi}{N}kn}$$

Analysis (Forward DFS):

$$\tilde{X}[k] = \frac{1}{N} \sum_{n=0}^{N-1} \tilde{x}[n] e^{-j\frac{2\pi}{N}kn}$$

2. Properties of the DFS

- Periodicity
- Linearity
- Shifting
- Symmetry
- Parseval' s Theorem
- Periodic/Circular Convolution

(1) Periodicity

Time Domain: The signal $\tilde{x}[n]$ is periodic with period N :

$$\tilde{x}[n + N] = \tilde{x}[n] \quad \forall n$$

Frequency Domain: The DFS coefficients $X[k]$ are also periodic with period N :

$$\tilde{X}[k + N] = \tilde{X}[k] \quad \forall k$$

Implication: Both the signal and its spectrum repeat every N samples

(2) Linearity

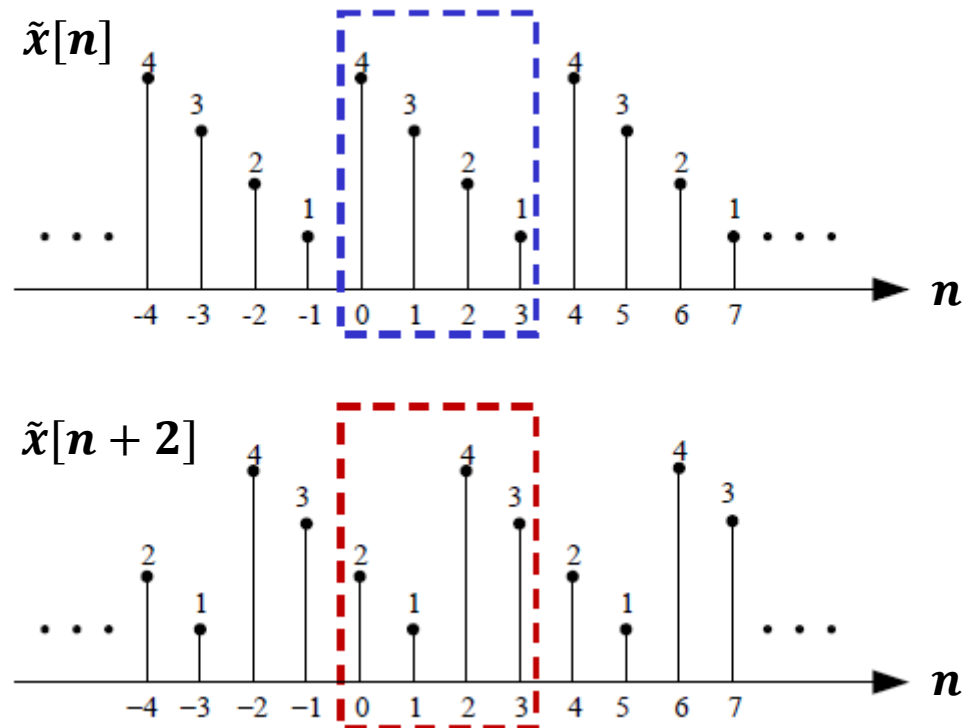
For two periodic signals $\tilde{x}_1[n]$ and $\tilde{x}_2[n]$ with DFS coefficients $\tilde{X}_1[k]$ and $\tilde{X}_2[k]$, and constants a and b :

$$a\tilde{x}_1[n] + b\tilde{x}_2[n] \xrightarrow{DFS} a\tilde{X}_1[k] + b\tilde{X}_2[k]$$

Implication: DFS preserves linear combinations, making it useful for superposition-based analysis

6. Discrete Time Fourier Series

(3) Shifting



A shift in the time domain still results in a periodic sequence with the same period. Thus, it is sufficient to analyze the shifted sequence over one period

(3) Shifting

Time Shifting

Shifting the signal by n_0 samples modulates the DFS coefficients:

$$\tilde{x}[n - n_0] \leftrightarrow \tilde{X}[k]e^{-j\frac{2\pi}{N}kn_0}$$

Implication: A **delay in time** introduces a linear **phase shift** in the **frequency domain**

(3) Shifting

Frequency Shifting

Modulating the signal by a complex exponential shifts its spectrum:

$$\tilde{X}[k - k_0] \leftrightarrow \tilde{x}[n] e^{j\frac{2\pi}{N}k_0n}$$

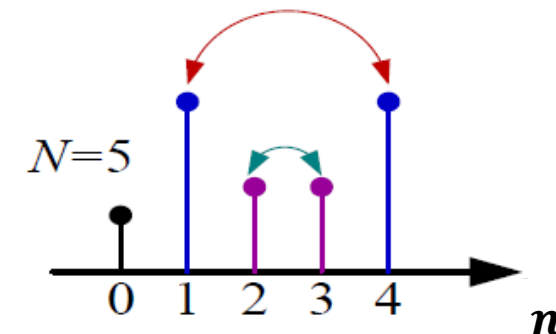
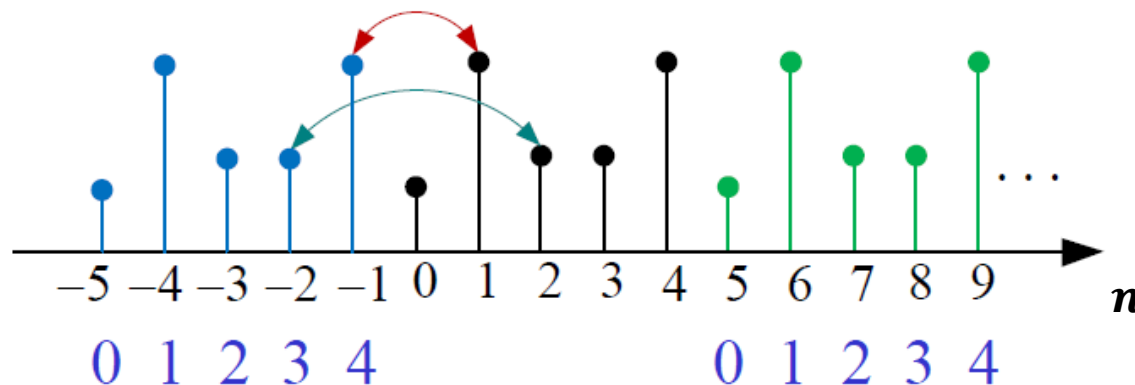
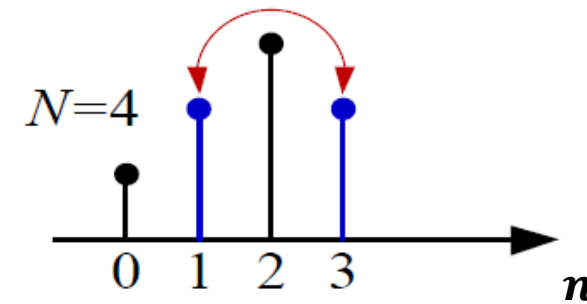
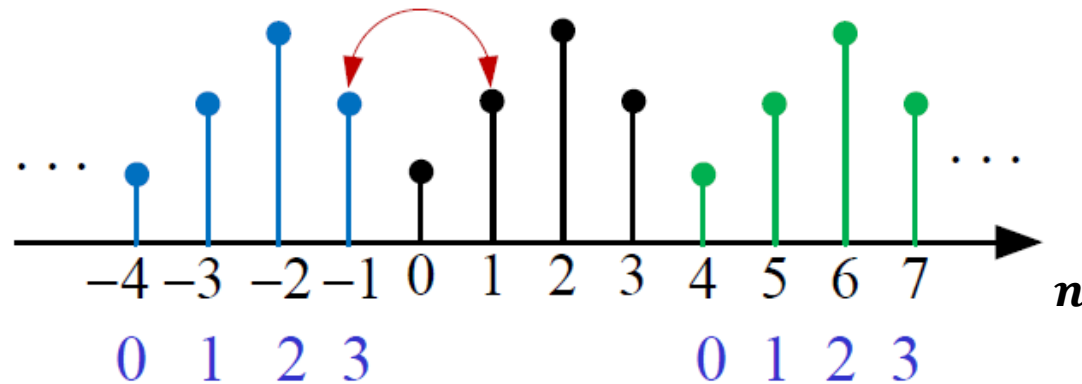
A modulation in the time-domain corresponds to a frequency-domain displacement

Implication: Used in modulation techniques (e.g., AM/FM)

6. Discrete Time Fourier Series

(4) Symmetry

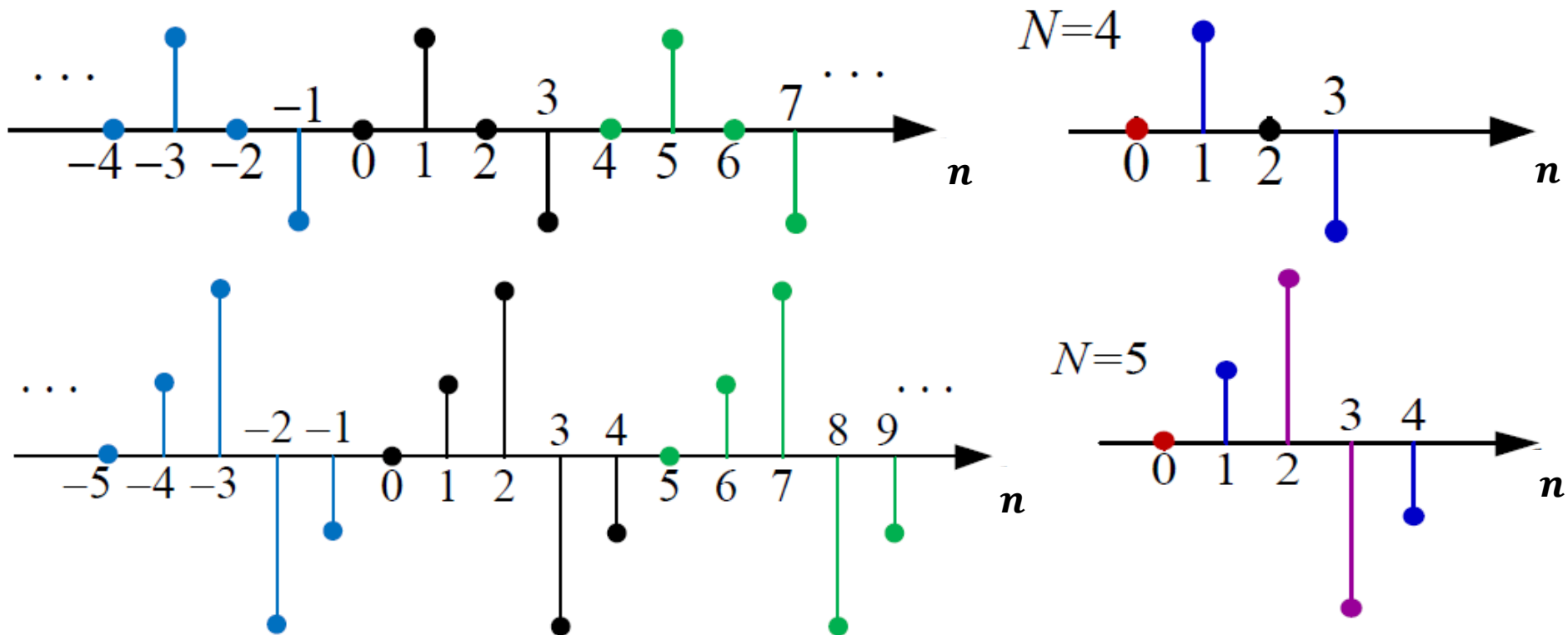
Even Symmetry in
Periodic Sequences : $\tilde{x}[n] = \tilde{x}[-n] = \tilde{x}[N - n]$



6. Discrete Time Fourier Series

(4) Symmetry

Odd Symmetry in
Periodic Sequences : $\tilde{x}[n] = -\tilde{x}[-n] = -\tilde{x}[N - n]$



(4) Symmetry

If

$$\tilde{x}[n] \xleftrightarrow{\text{DFS}} \tilde{X}[k],$$

Then

1. Conjugation in time:

$$\tilde{x}^*[n] \xleftrightarrow{\text{DFS}} \tilde{X}^*[-k]$$

2. Conjugation & reversal in time:

$$\tilde{x}^*[-n] \xleftrightarrow{\text{DFS}} \tilde{X}^*[k]$$

(5) Parseval's Theorem

Relates the total **energy** in the time and frequency domains for one N -point period:

$$\sum_{n=0}^{N-1} |\tilde{x}[n]|^2 = N \sum_{k=0}^{N-1} |\tilde{X}[k]|^2$$

Implication: Energy is conserved between domains (up to scaling factor N), and $|\tilde{X}[k]|^2$ represents the energy at frequency bin k , $1/N |\tilde{X}[k]|^2$ represents the **power** at frequency bin k

(6) Circular Convolution

Circular Convolution: **Multiplication** in one domain corresponds to **circular convolution** in the other:

$$\tilde{x}[n] \circledast \tilde{h}[n] \xrightarrow{DFS} N \cdot \tilde{X}[k] \cdot \tilde{H}[k]$$

$$\tilde{x}[n] \cdot \tilde{h}[n] \xrightarrow{DFS} \tilde{X}[k] \circledast \tilde{H}[k]$$

Here, $\circledast$ denotes circular convolution

Implication: Efficient for filtering periodic signals (e.g., in digital communications)

Main Content

- Discrete Time Fourier Transform (DTFT)
- Discrete Fourier Transform (DFT)
- Properties of DFT

1. Discrete Time Fourier Transform (DTFT)

Start with the Fourier Transform for Continuous Signals: The continuous-time Fourier Transform (CTFT) is:

$$X(j\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t} dt$$

For discrete-time signals, replace t with nT (sampling interval T) and the integral with a sum:

$$X(e^{j\Omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\Omega n}$$

1. Discrete Time Fourier Transform (DTFT)

The DTFT of a sequence $x[n]$ is given by:

$$X(e^{j\Omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\Omega n}$$

where:

- Ω is the normalized angular frequency ($\Omega = 2\pi f$)
- $X(e^{j\Omega})$ is periodic with period 2π

Inverse DTFT (IDTFT):

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\Omega}) e^{j\Omega n} d\Omega$$

1. Discrete Time Fourier Transform (DTFT)

- **Periodicity of DTFT:** Since $e^{-j\Omega n} = e^{-j(\Omega+2\pi)n}$, the DTFT is periodic with 2π
- **Convergence Conditions:** The DTFT exists if $x[n]$ is absolutely summable:

$$\sum_{n=-\infty}^{\infty} |x[n]| < \infty$$

2. Discrete Fourier Transform (DFT)

Deriving the DFT from the DTFT

Step 1: Truncate the Signal

The DTFT assumes an infinite-length sequence. To make it practical:

- Consider a finite-length sequence $x[n]$ defined for $n = 0, 1, \dots, N - 1$
- For all other n , $x[n] = 0$

Step 2: Sample the DTFT in Frequency

The DTFT $X(e^{j\Omega})$ is continuous in Ω . To discretize it:

- Evaluate $X(e^{j\Omega})$ at equally spaced frequencies:

$$\Omega_k = \frac{2\pi k}{N}, k = 0, 1, \dots, N - 1$$

- Substituting into the DTFT:

$$X[k] = X(e^{j\Omega_k}) = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}kn}$$

**Frequency
domain sampling**

2. Discrete Fourier Transform (DFT)

Transform	Domain	Frequency Representation	Key Characteristics
DTFT	Infinite-length sequences	Continuous ($\Omega \in [-\pi, \pi]$)	$X(e^{j\Omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\Omega n}$
DFT	Finite-length sequences (N samples)	Discrete ($k = 0, 1, \dots, N - 1$)	$X[k] = \sum_{n=0}^{N-1} x[n]e^{-j\frac{2\pi}{N}kn}$

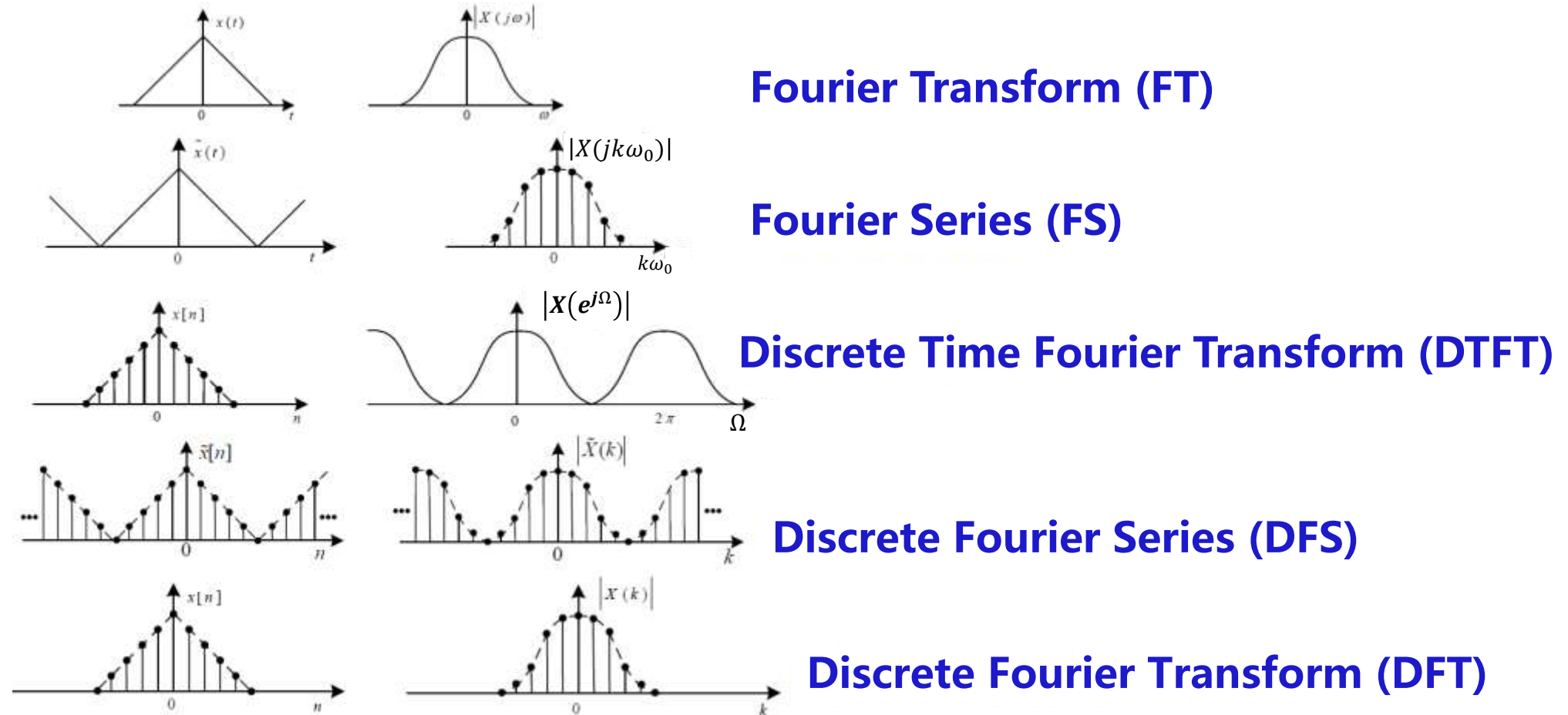
7. DTFT and DFT

Fourier Transform Family: Complete Comparison

Transformation	Time domain characteristics	Frequency domain characteristics
Fourier Transform (FT)	Continuous, Aperiodic $x(t) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(j\omega) e^{j\omega t} d\omega$	Continuous, Aperiodic $x(j\omega) = \int_{-\infty}^{+\infty} x(t) e^{-j\omega t} dt$
Fourier series (FS)	Continuous, Periodic $\tilde{x}(t) = \sum_{k=-\infty}^{\infty} X(jk\omega_0) e^{jk\omega_0 t}$	Discrete, Aperiodic $X(jk\omega_0) = \frac{1}{T} \int_0^T \tilde{x}(t) e^{-jk\omega_0 t} dt$
Discrete-time Fourier transform (DTFT)	Discrete, Aperiodic $x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\Omega}) e^{j\Omega n} d\Omega$	Continuous, Periodic $X(e^{j\Omega}) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\Omega n}$
Discrete Fourier series (DFS)	Discrete, Periodic $\tilde{x}[n] = \sum_{k=0}^{N-1} \tilde{X}(k) e^{j\frac{2\pi}{N}kn},$ $-\infty < n < +\infty$	Discrete, Periodic $\tilde{X}[k] = \frac{1}{N} \sum_{n=0}^{N-1} \tilde{x}(n) e^{-j\frac{2\pi}{N}kn},$ $-\infty < k < +\infty$
Discrete Fourier transform (DFT)	Discrete, Implicit Periodic $x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X(k) e^{j\frac{2\pi}{N}kn},$ $0 \leq n \leq N-1$	Discrete, Implicit Periodic $X[k] = \sum_{n=0}^{N-1} x(n) e^{-j\frac{2\pi}{N}kn},$ $0 \leq k \leq N-1$

7. DTFT and DFT

Fourier Transform Family: Complete Comparison



DFT is DFS evaluated over a single period (**principal value interval**) of a finite-length sequence

2. Discrete Fourier Transform (DFT)

The definition formula for the discrete Fourier transform of a finite time domain (length N) discrete time series $x(n)$ is as follows:

$$\begin{aligned} X(k) &= \text{DFT}[x(n)] \\ &= \sum_{n=0}^{N-1} x(n) e^{-j\frac{2\pi}{N}nk} \quad 0 \leq k \leq N-1 \end{aligned} \quad W = e^{-j\frac{2\pi}{N}} \quad X(k) = \sum_{n=0}^{N-1} x(n) W^{nk}$$

The corresponding inverse transformation of discrete Fourier transform is

$$\begin{aligned} x(n) &= \text{IDFT}[X(k)] \\ &= \frac{1}{N} \sum_{k=0}^{N-1} X(k) e^{j\frac{2\pi}{N}nk} \quad 0 \leq n \leq N-1 \end{aligned} \quad x(n) = \frac{1}{N} \sum_{k=0}^{N-1} X(k) W^{-nk}$$

k is the index on the frequency domain, while n is the index on the time domain, with N is the length of sequence $0 \sim (N-1)$

W is the **twiddle** factor (**phase rotation** factor)

2. Discrete Fourier Transform (DFT)

$$\begin{bmatrix} X(0) \\ X(1) \\ \vdots \\ X(N-1) \end{bmatrix} = \begin{bmatrix} W^0 & W^0 & W^0 & \dots & W^0 \\ W^0 & W^{1 \times 1} & W^{2 \times 1} & \dots & W^{(N-1) \times 1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ W^0 & W^{1 \times (N-1)} & W^{2 \times (N-1)} & \dots & W^{(N-1) \times (N-1)} \end{bmatrix} \begin{bmatrix} x(0) \\ x(1) \\ \vdots \\ x(N-1) \end{bmatrix}$$

$$\begin{bmatrix} x(0) \\ x(1) \\ \vdots \\ x(N-1) \end{bmatrix} = \frac{1}{N} \begin{bmatrix} W^0 & W^0 & W^0 & \dots & W^0 \\ W^0 & W^{-1 \times 1} & W^{-2 \times 1} & \dots & W^{-(N-1) \times 1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ W^0 & W^{-1 \times (N-1)} & W^{-2 \times (N-1)} & \dots & W^{-(N-1) \times (N-1)} \end{bmatrix} \begin{bmatrix} X(0) \\ X(1) \\ \vdots \\ X(N-1) \end{bmatrix}$$

Example 1

1. Calculate the DFT of a finite length sequence $\{h(n)\} = \{1, 2, -1, 3\}$

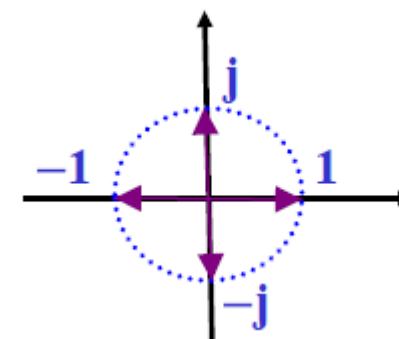
Solve: $H(k) = \sum_{n=0}^3 h(n) W_4^{nk}, \quad N = 4$

$$\begin{bmatrix} H(0) \\ H(1) \\ H(2) \\ H(3) \end{bmatrix} = \begin{bmatrix} W_4^{0 \cdot 0} & W_4^{0 \cdot 1} & W_4^{0 \cdot 2} & W_4^{0 \cdot 3} \\ W_4^{1 \cdot 0} & W_4^{1 \cdot 1} & W_4^{1 \cdot 2} & W_4^{1 \cdot 3} \\ W_4^{2 \cdot 0} & W_4^{2 \cdot 1} & W_4^{2 \cdot 2} & W_4^{2 \cdot 3} \\ W_4^{3 \cdot 0} & W_4^{3 \cdot 1} & W_4^{3 \cdot 2} & W_4^{3 \cdot 3} \end{bmatrix} \begin{bmatrix} h(0) \\ h(1) \\ h(2) \\ h(3) \end{bmatrix}$$

$$W_4^1 = e^{-j\frac{2\pi}{4}} = -j, \quad W_4^2 = -1, \quad W_4^3 = W_4^1 = W_4^2 = j, \quad W_4^4 = W_4^0 = 1$$

$$\begin{bmatrix} H(0) \\ H(1) \\ H(2) \\ H(3) \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ -1 \\ 3 \end{bmatrix} = \begin{bmatrix} 5 \\ 2 + j \\ -5 \\ 2 - j \end{bmatrix}$$

$$\{H(k)\} = \{5, 2 + j, -5, 2 - j\}$$



3. Properties of DFT

- **Linearity**
- **Circular (Periodic) Shift**
- **Circular Convolution**

(1) Linearity

If the linear combination of finite length sequences $x_1(n)$ and $x_2(n)$ is

$$x_3(n) = ax_1(n) + bx_2(n)$$

a, b : constants

$X_1(k)$, $X_2(k)$ and $X_3(k)$ are DFTs of $x_1(n)$, $x_2(n)$ and $x_3(n)$ respectively

$$\text{Then, } X_3(k) = aX_1(k) + bX_2(k)$$

If the lengths of $x_1(n)$ and $x_2(n)$ are N_1 and N_2 respectively, with $N_2 > N_1$, the shorter sequence $x_1(n)$ must be **zero-padded** by appending L zeros to match the length of $x_2(n)$. This ensures $N_2 = N_1 + L$, the resulting sequence $x_3(n)$ will then have a length of N_2

(2) Circular (Periodic) Shift

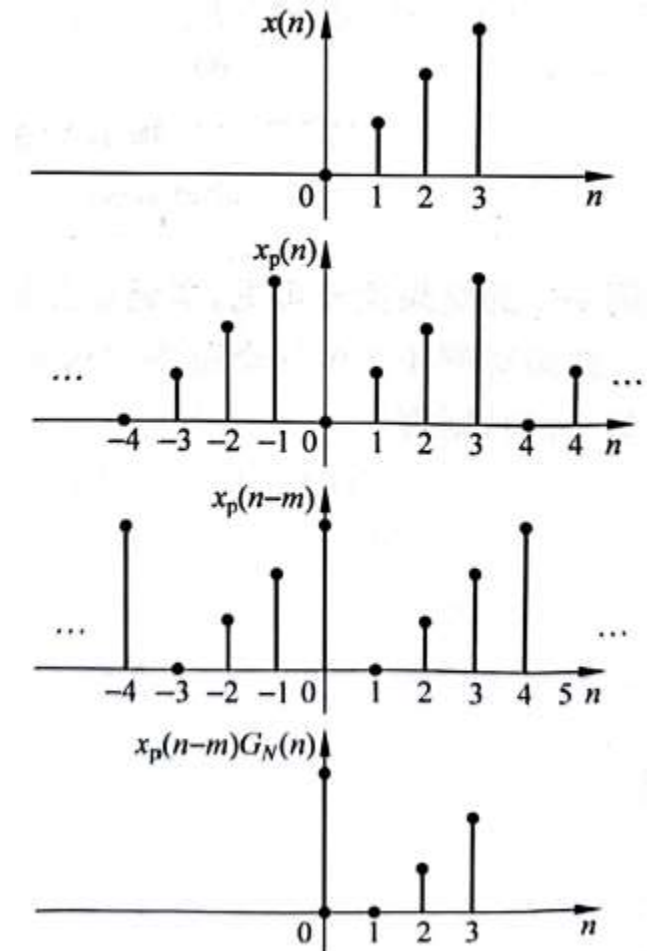
Periodic Extension:

The sequence $x(n)$ with a length of N is periodically extended to generate the periodic sequence $x_p(n)$

Circular Shift by m Samples

$$x_p(n - m)$$

Principal Value Interval Extraction



(2) Circular (Periodic) Shift

Principal Value Interval Extraction

$$x_p(n) = \sum_{r=-\infty}^{\infty} x(n - rN), \quad r \text{ is an integer}$$

$$\text{and, } 0 \leq n - rN \leq N - 1$$

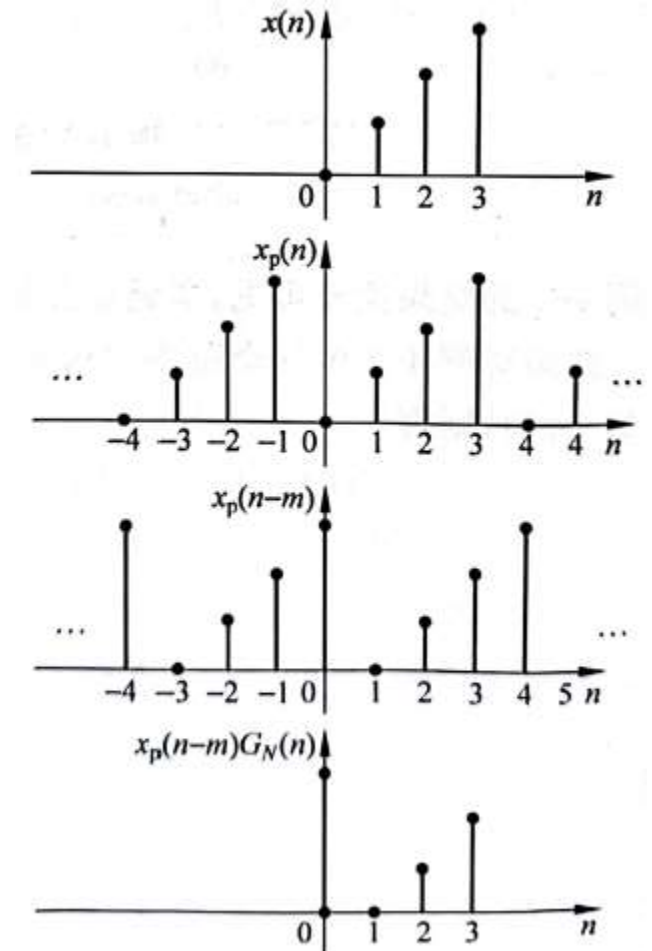
Define a rectangular window

$$G_N(n) = u(n) - u(n - N)$$

Then

$$x(n) = x_p(n)G_N(n)$$

$x_p(n \pm m) \cdot G_N(n)$ represents the sequence obtained by taking the principal interval of a periodic signal after a circular shift of $\pm m$, with a length of N



(2) Circular (Periodic) Shift

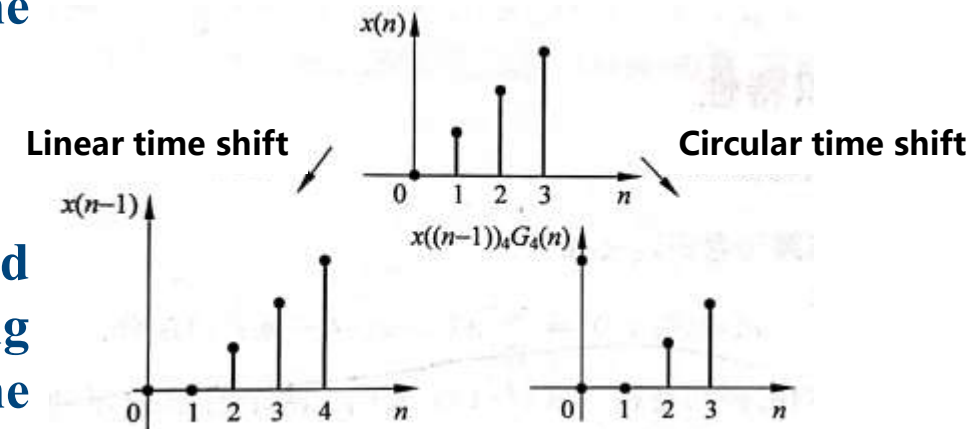
Linear time shift vs. **circular** time shift

Circular time shift:

The interval remains unchanged, the positions shift, and the magnitudes before and after shifting remain the same

Sequential circular shift by m samples:

The N samples of $x(n)$ are arranged around a circle divided into N equal parts (with the start and end connected). Rotating the sequence by m positions around the circle produces the circular shift



$$\{x(n)\} = \{\mathbf{1}, 0, 2, 3\}$$

$$\{x(n-1)\} = \{0, \mathbf{1}, 0, 2, 3\}$$

$$\{x_c(n)\} = \{x_c((n-1))_4 G_4(n)\} = \{3, \mathbf{1}, 0, 2\}$$

(2) Circular (Periodic) Shift

If

$$\text{DFT}\{x(n)\} = X(k),$$

then shifting $x(n)$ circularly by m samples gives:

$$\text{DFT}\{x((n - m))_N\} = W_N^{+mk} X(k),$$

where

$$X(k) = \sum_{n=0}^{N-1} x(n) W_N^{nk}, W_N = e^{-j\frac{2\pi}{N}}$$

Usage?

(2) Circular (Periodic) Shift

If

$$\text{DFT}\{x(n)\} = X(k),$$

then a circular shift of the spectrum by m :

$$X_c(k) = X((k - m))_N$$

corresponds in the time domain to:

$$x_c(n) = W_N^{-mn} x(n),$$

where

$$W_N = e^{-j\frac{2\pi}{N}}$$

Usage?

(3) Circular Convolution

Time domain circular convolution of two sequences

$$x(n) \circledast h(n) = \sum_{m=0}^{N-1} x(m)h((n-m))_N G_N(n)$$

$\circledast$ Circular Convolutions Representing Two Sequences

Periodically extending and circularly shifting one sequence $h(m)$ with period N , multiplying it with the other sequence $x(m)$, summing the products

The length of sequences $x(m)$ and $h(m)$ should both be N , and the entire convolution process is carried out in the $0 \leq m \leq (N-1)$ interval, with the length $L = N$ after convolution

$((n-m))_N$ denotes $(n-m)$ modulo N , representing the circular shift of the sequence $h(m)$

Example 1

Two finite length sequences are known

$$(N=4)$$

$$x(n) = \delta(n) + \delta(n-1) + \delta(n-2)$$

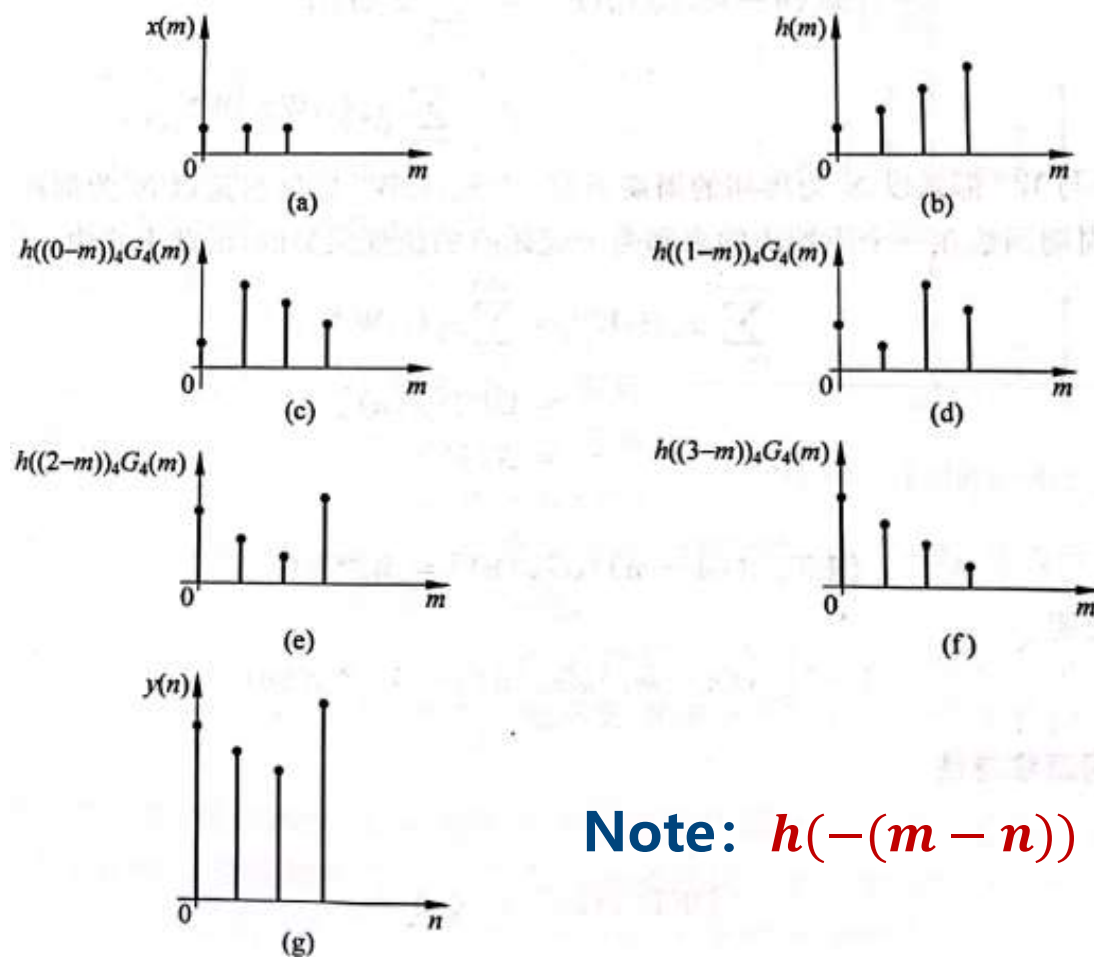
$$h(n) = (n+1)G_4(n)$$

Graphical method for solving circular convolution

$$y(n) = x(n) \circledast h(n)$$

(3) Circular Convolution

Replace the independent variables n of $x(n)$ and $h(n)$ with m



$$x(n) \circledast h(n) = \sum_{m=0}^{N-1} x(m)h((n-m))_N G_N(n)$$

$$y(n) = \sum_{m=0}^3 x(m)h((n-m))_4 G_4(n)$$

$$y(0) = 1 \times 1 + 1 \times 4 + 1 \times 3 + 0 \times 2 = 8$$

$$y(1) = 1 \times 2 + 1 \times 1 + 1 \times 4 + 0 \times 3 = 7$$

$$y(2) = 1 \times 3 + 1 \times 2 + 1 \times 1 + 0 \times 4 = 6$$

$$y(3) = 1 \times 4 + 1 \times 3 + 1 \times 2 + 0 \times 1 = 9$$

obtain,

$$y(n) = 8\delta(n) + 7\delta(n-1) + 6\delta(n-2) + 9\delta(n-3)$$

Note: $h(-(m-n))$

(3) Circular Convolution

Linear convolution vs Circular convolution

$$y(n) = x(n) * h(n) = \sum_{m=-\infty}^{\infty} x(m)h(n-m)$$

$$y(n) = x(n) \circledast h(n) = \sum_{m=0}^{N-1} x(m)h((n-m))_N G_N(n)$$

(3) Circular Convolution

1. Linear Convolution:

- Sequences $x(n)$ [length N] and $h(n)$ [length M] can have different lengths
- Resulting sequence length: $L = N + M - 1$
- Represents normal time-domain filtering operations

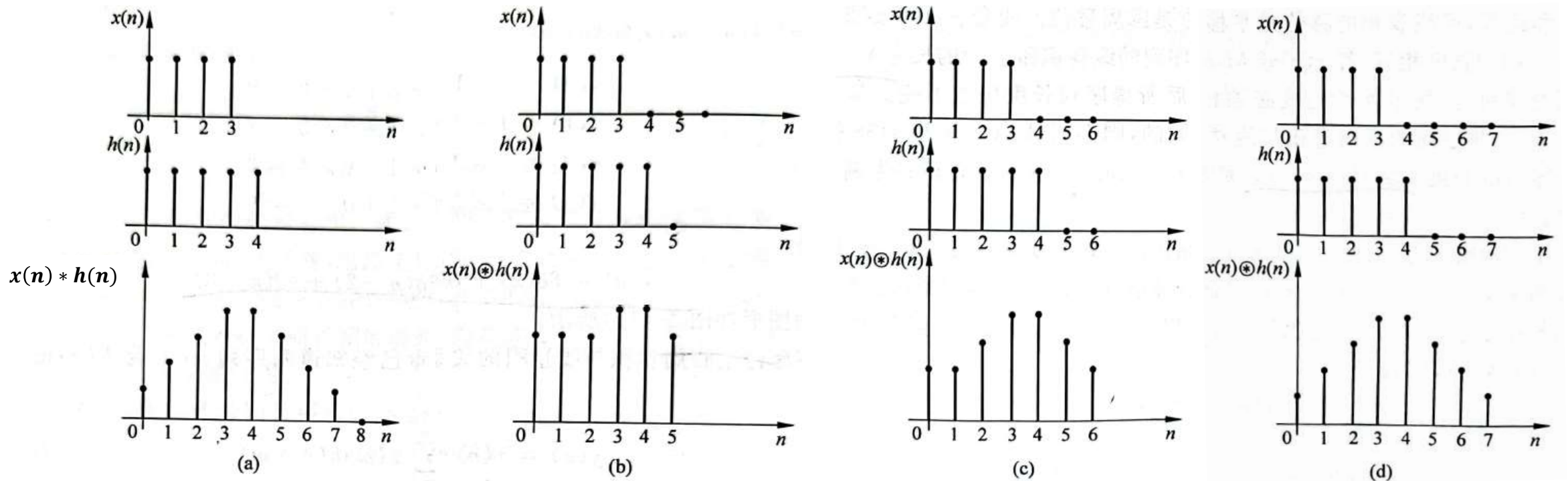
2. Circular Convolution

- Requires both sequences to have equal lengths
- Output sequence maintains the same length as inputs
- Fundamental to DFT operations

To make circular convolution produce **identical results** to linear convolution:

1. Both sequences must be zero-padded to length $L \geq N + M - 1$
2. The padded length L must be identical for both sequences

(3) Circular Convolution



(a) Linear convolution; (b) Circular convolution with $L=6$; (c) Circular Convolution with $L=7$; (d) Circular Convolution with $L=8$

$$L = N + M - 1 = 8$$

$$x(n) * h(n) = x(n) \circledast h(n)$$

(3) Circular Convolution

If

$$\text{DFT}\{x(n)\} = X(k), \text{DFT}\{h(n)\} = H(k),$$

then the circular convolution of $x(n)$ and $h(n)$:

$$y(n) = x(n) \circledast h(n),$$

has the DFT

$$Y(k) = \text{DFT}\{y(n)\} = X(k) H(k)$$

Circular convolution in time corresponds to multiplication in the frequency domain

(3) Circular Convolution

If

$$\text{DFT}\{x(n)\} = X(k), \text{DFT}\{h(n)\} = H(k),$$

and

$$y(n) = x(n) \cdot h(n), Y(k) = \text{DFT}\{y(n)\},$$

then

$$Y(k) = \frac{1}{N} [X(k) \circledast H(k)],$$

where $\circledast$ denotes circular convolution in the frequency domain

Multiplication in the time domain corresponds to circular convolution in the frequency domain (with scaling $1/N$)



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Thank you for listening !

